



Global Underwater Drone Market Study, Opportunities, & Forecast





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1 Introduction

1.1 Executive Summary

		2018	2025
Market Size (USD Million)		2,942	7,102
Growth Rate (CAGR 2018-2025)		13.4%	
Major Market Player		Oceaneering International, Inc.	
Geographical Segment	North America	1,179	2,799
	Europe	838	2,010
	Asia-Pacific	675	1,723
	Rest of the World	249	570
Key Market Segments	Remotely Operated Vehicle	2,099	4,478
	Commercial Exploration	1,505	3,730
Market Representation		Value in USD Million and Growth Rates in %	

Source: ERC Analysis, Government Publications, and Secondary Research

1.2 Market Overview

Under water drones, also called as unmanned underwater vehicles (UUV), are the vehicles which are capable of conducting underwater operations without human presence. These vehicles are either, remotely operated, which are directed by a remote human operator or fully autonomous, which can work without any human control, like a robot. The underwater drones have already been used by the military and defense authorities since a long time, but in the current scenario, the application platform of underwater drones is constantly expanding. Under water drones are well equipped with various types of cameras as per the requirement. A drastic usage of these drones has been witnessed among the oceanographers and filmmakers for making marine documentaries and other films. Today, everyone is focused on smart, quick and cheap way to execute even the most complicated tasks. Major market players such as Aquabotix has now come up with latest drone models which could be used by the fishermen in order to detect the actual location of shoals of fish. In addition to this, the constantly rising exploration activities in order to access natural oil and gas reservoirs is fueling the demand for underwater drones. Underwater drones have emerged out as an effective mode of getting underwater condition of rigs and monitoring the drilling processes. However, the high cost of camera equipped ROVs might act as a major challenge for the market. In terms of Geography, North America is expected to lead the global Underwater Drones Market owing to heavy R&D investment by military organizations and constantly rising Oil and Gas exploration activities as a result of shale boom in the region.

1.3 Market Scope

Global Underwater Drone market has been segmented majorly on the basis of Type, Application, Component, Propulsion System, and region. Geographically, global Underwater Drone market is divided as North America, Europe, Asia-Pacific, and Rest of the World.

By Type

- Remotely Operated Vehicle (ROV)
- Autonomous Underwater Vehicle (AUV)

By Application

- Commercial Exploration
- Defense
- Scientific Research
- Others

By Component

- Sensors
- Synthetic Aperture Sonar
- Inertial Navigation Systems
- Imaging Systems
- Lighting Systems
- Others

By Propulsion System

- Electric System
- Mechanical System
- Hybrid Systems

By Geography

- North America
 - U.S.
 - Canada
 - Mexico
- Europe
 - Germany
 - France
 - UK
 - Italy
 - Spain
 - Rest of Europe
- Asia-Pacific
 - China
 - Japan
 - India
 - Australia
 - Rest of Asia-Pacific
- Rest of the World
 - Brazil
 - Argentina
 - Rest of Rest of the World
- Middle East and Africa
 - GCC Countries
 - Egypt
 - South Africa
 - Rest of Middle East & Africa

2 Research Methodology

2.1 Primary Research

We at Esticast Research & Consulting employ iterative research methodology along with a combination of secondary and primary research to provide the most accurate market estimates and forecast. Key research steps include

- Data mining

This step includes in-depth data mining wherein we procure market data as well as any related information through different verified databases and sources.

- Data analysis

The procured information is then analyzed and mapped to assess the data as well as eliminate any discrepancies observed.

- Market Modeling

All the gathered data points are then studied to devise appropriate estimation and sizing model in an attempt to deduce viable conclusions. Subject-matter-expert and analyst perspective & based heuristic form of market sizing is an integral part of this stage

- Validation & publishing

Validation is the most important step in the process. Validation & revalidation via an intricately designed process helps us finalize data points to be used for final calculations.

2.2 Research Methodology

We implemented a combination of primary and secondary research to estimate and size our market. Secondary research was the initial phase involved in our study. This is where we conducted data mining and referred to verified data sources, such as independent studies, government and regulatory published material, technical journals, trade magazines, and paid data sources.

For forecasting, the following parameters were considered:

- Market drivers, challenges and opportunities along with their current and expected impacts
- Technological developments and current status with expected developments
- Products types and application trends and dynamics
- Consumer trends
- Unmet need analysis and demand forecasting

Weights were assigned to these parameters and their impact on the market was quantified using a weighted average technique in order to derive expected market growth rate.

All the estimates and forecasts are verified through extensive primary research with experts operating in this market, and include:

- Key market participants
- Manufacturers
- Sales and marketing team

2.3 Assumptions & Exclusions

- The report provides market size with 2016 as the base year in consideration and a yearly forecast till 2025 in terms of Revenue (USD Million). The estimates for all segments including Type, and application have been provided on a regional basis for the forecast period mentioned above
- We have implemented a mix of top-down and bottom-up approach for market sizing, analyzing the key regional markets, dynamics, and trends for various applications. The Global market has been estimated by integrating the regional markets.
- **Market estimates have mainly been based on the feedback obtained through primary interviews with key industry participants**
- Inflation has not been accounted for in order to estimate the market
- Numbers may not add up due to rounding off

2.4 Secondary data sources

List of secondary sources includes but is not limited to:

- Hoover's
- Factiva
- Consumer Electronics Association (CEA)
- Secondary Research
- PC Magazines
- Physics Forum
- Investor Presentations
- Company Annual Reports

3 Market Dynamics

3.1 Market Drivers

3.1.1 Growth in demand across end-user verticals

The demand for ROVs and AUVs is continuously increasing due to the variety of applications across the oil and gas, scientific research, and defense industries. In the defense industry, ROVs are used for counterattacking and for deactivating underwater mines, as well as for port security and hull inspection. The increased usage of ROVs in defense operations such as reconnaissance missions and mine countermeasures will drive the market in the future. ROVs and AUVs also have many applications in scientific research, especially oceanographic studies, where they are used to collect undersea data and map and study aquatic life. They are widely used because of their cost-effectiveness, reliability, rapid deployment, and safer operational capabilities. Globally, the increase in demand for energy will drive the oil and gas sector. AUVs are extensively used, as they can perform complex tasks in oilrig construction, pipeline inspection, and maintenance. Energy consumption globally has tripled over the last five decades due to the growth of developing economies. The global annual oil consumption was 99.84M bbl./d for 2018, projected to increase to 100.6 M bbl./d in 2019. Thus, the continuous rise in the demand for energy will propel the growth prospects of the Underwater Drones market.

3.1.2 Rise in offshore construction and drilling activities

The increase in offshore construction and drilling services would significantly influence the growth of underwater drone market during the forecast period, the result of increased usage of oil worldwide. Work class ROVs play an essential role in offshore construction because they are useful for deep-water oil production. According to ERC analysis and secondary data, U.S. drilling activities increased by 27.1% in 2017, to 18,552 operational wells. The U.S. Gulf of Mexico E&P activity focused on deep-water projects, went up approximately 9.5% in 2017, with increasing well depths and footage. Global drilling increased to 39,742 wells, for a 6.1% pick-up in the same year. The product benefits of ROVs and AUVs with regards to deep-water oil & gas exploration, such as extended dive duration, quick vehicle deployment, and minimal maintenance, would fuel the growth in demand for underwater drones, boosting overall market growth.

3.2 Market Restraints

3.2.1 Issues and complexities during drone operation

Underwater drones are controlled by either software or a human operator. They are equipped with different payloads, sensors, manipulators, and cameras, which are used to perform a variety of functions. However, the systems might face problems owing to software crashes and sensor failures, resulting in the loss of AUVs at sea. In addition, water limits the transmission of visible light and electromagnetic waves; the inability of electromagnetic waves to transmit through water impedes operators' observation of the sea and results in communication problems underwater. The difficulties with teleoperation concern ROVs, which require manual control, while complexities with supervisory control are persistent with AUVs. In the latter case, the operator faces challenges in communicating with the AUV and in observing the underwater environment, as the operator is dependent on a video camera. Poor vision limits the operator's ability to perform multiple tasks, including the ability to maneuver the vehicle, detect objects in the environment, and navigate or handle objects. Poor vision leads to accidents resulting in damage to the robot, increased costs, and lost time. The damage in property leads to additional expenditure and spending, acting as limitations to extensive product adoption.

3.3 Market Opportunities

3.3.1 Rapid technological development to increase product capabilities

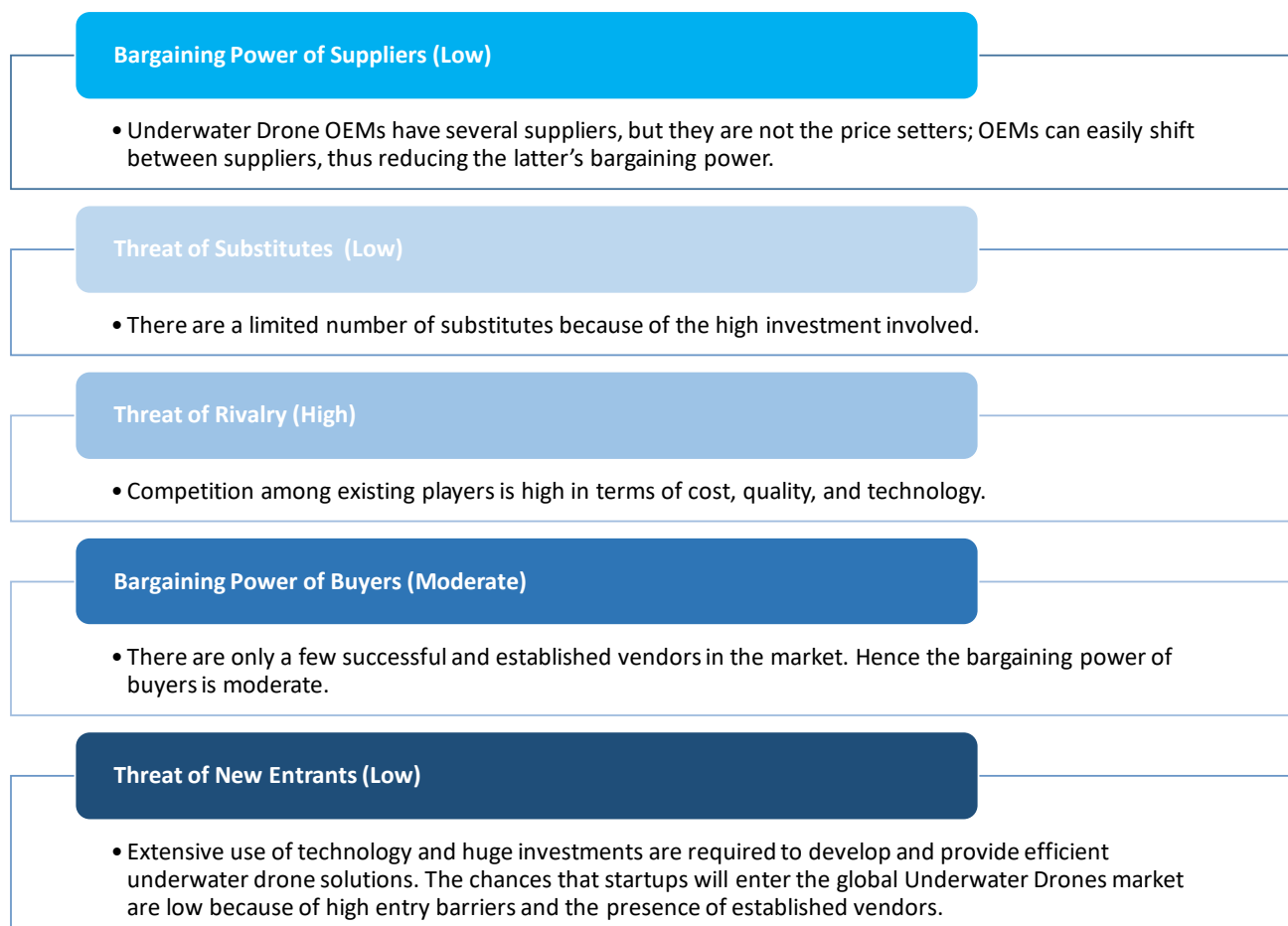
Advances in technology have led to a decrease in the size of ROVs. Small ROVs are preferred to large ROVs because of their low deployment cost, better maneuverability, and lower maintenance costs. Small ROVs, once deployed, require fewer personnel to operate, while large ROVs require several operators. The introduction of hybrid ROVs without tethered cables have become an optional configuration. In such a configuration, the vehicle can perform programmed missions free from cables. The significant benefit of these hybrid systems is greater maneuverability in places where a tether would otherwise restrict the operator's ability to navigate around obstacles. The product development of AUVs has also resulted in additional capabilities. AUVs are currently utilized in the military to trace and disable mines. These AUVs have to travel long distances to locate the mines, and hence there is a continuous need to increase the mission life of AUVs. Current AUVs can operate for up to one year, after which batteries have to be replaced and the vehicles reconditioned. Thus, enhanced battery life will increase the mission life of AUVs. Compact batteries are one technology that can increase the mission life of AUVs; they have the ability to store more electricity than other batteries. In addition, improved communication and instrumentation devices will allow AUVs to consume less power. Alternatively, the use of solar arrays kept above the surface of the ocean, or bio-mimicking technology can generate onboard power.

3.4 Porter's Five Forces Analysis

Porter's Five Forces analysis of the Underwater Drone market considers both the micro and macro environmental factors prevailing in the market.

3.4.1 Porter's Five Forces Analysis

FIGURE 1 BARGAINING POWER OF SUPPLIERS



Source: ERC Analysis, Government Publications, and Secondary Research

3.5 Competitive Landscape

The global Underwater Drones market witness intensive competitiveness among its key players looking to expand business activities and strengthen business position on the basis of innovative product capabilities and R&D development. Design and manufacturing capabilities using new technologies and innovation in the field of unmanned vehicles by companies with a global approach and vast clientele has created competition in the market.

The underwater drone solutions have gained much popularity in commercial industries for drilling support, subsea development, and scientific research, prompting vendors to innovate and upgrade with their clients' unique needs in mind. The global Underwater Drones market is thus highly competitive, and vendors compete on the basis of cost, quality, reliability, and aftermarket service. To succeed in such an intensely competitive environment, it is crucial for vendors to provide cost-effective and high-quality service and the latest technology.

Continued focus on product development and formation of strategic partnerships and collaborations are the most followed strategies of key manufacturers and market vendors, intending to maintain competitive advantage over other players. For instance, on July 05, 2018, Kawasaki Heavy Industries, Ltd. announced that a basic agreement has been reached with The Underwater Centre (TUC), a marine testing and training facility in Fort William, Scotland, UK, on conducting a verification test of a prototype AUV equipped with a robot arm for subsea pipeline inspection. Further, on June 08, 2017, The Boeing Company and Huntington Ingalls announced a partnership to develop a cost-effective, low-risk extra-large unmanned undersea vehicles as part of the Navy's Advanced Undersea Prototyping program. Moreover, on September 26, 2017, Hydroid, Inc. announced that it has partnered with L3 Aerospace Systems, Sparton Corporation, and AeroVironment, Inc. to successfully demonstrate the latest unmanned underwater vehicle (UUV) capabilities, including the launch of an unmanned aerial vehicle (UAV), the Hydroid REMUS 600. Furthermore, on September 09, 2016, General Dynamics Bluefin-21 autonomous underwater vehicle (AUV) successfully launched multiple Bluefin SandShark micro-autonomous underwater vehicles (M-AUV) in the commercial market.

4 Type Overview

4.1 Introduction

Global Underwater Drone market is segmented on the basis of Type by Remotely Operated Vehicle and Autonomous Underwater Vehicle. ROV segment dominates the market share. However, AUV is the fastest growing segment of the global Underwater Drone market.

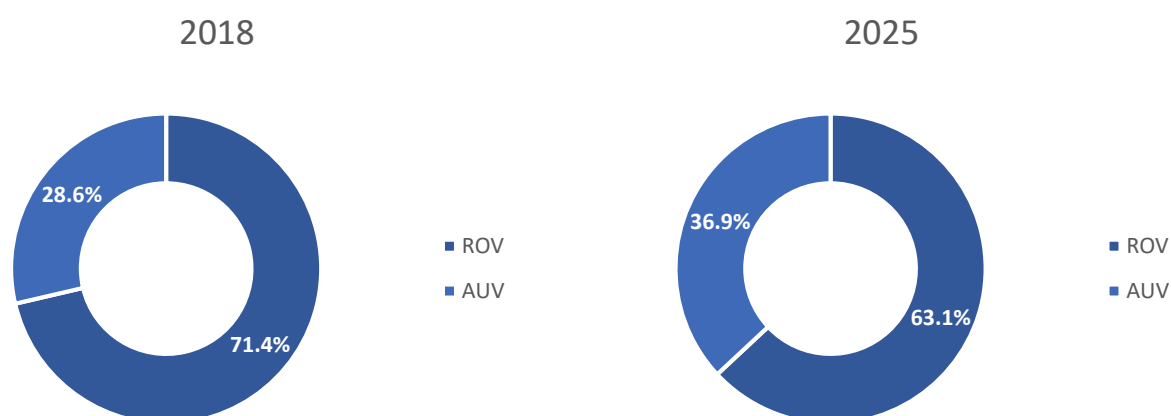
4.1.1 Market Size & Forecast

TABLE 1 GLOBAL UNDERWATER DRONE MARKET SHARE, BY TYPE, 2016 – 2025 (\$MILLION)

Type	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
ROV	1,694	1,885	2,099	2,337	2,602	2,899	3,230	3,601	4,015	4,478	11.4%
AUV	598	710	842	996	1,175	1,384	1,628	1,911	2,241	2,624	17.5%
Total	2,292	2,596	2,942	3,333	3,778	4,284	4,858	5,513	6,256	7,102	13.4%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

FIGURE 2 GLOBAL UNDERWATER DRONE MARKET SHARE BY TYPE, 2018 VS 2025 (%)



Source: ERC Analysis, Government Publications, and Secondary Research

4.2 Remotely Operated Vehicle

ROV Underwater Drone market segment is estimated to be \$ 2,099 million in 2018 and expected to garner \$ 4,478 million by 2025. North America is expected to be largest market for ROV Underwater Drone and is anticipated to grow at a CAGR of 11.3% from 2019-2025. ROV segment is projected to grow at a CAGR of 11.4% from 2019 to 2025.

A remotely operated vehicle (ROV) is an unoccupied underwater robot that is connected to a ship by a series of cables. ROVs are equipped with at least a still camera, video camera, and lights, meaning that they can transmit images and video back to the ship. The other additional equipment, such as a manipulator or cutting arm, water samplers, and instruments that measure parameters like water clarity and temperature, may also be added to vehicles to allow for sample collection. Remotely operated vehicles (ROVs) are a safe and widely used type of underwater vehicle serving a range of military, commercial, and scientific needs. ROVs are unoccupied, highly maneuverable underwater robots operated by a person on the surface. The ROV, allowing remote navigation of the vehicle.

The startup entering into underwater drone market and developing new products is boosting the market growth. For instance, on Mar 19, 2019, Planys Technologies, a startup providing underwater robotic inspection solutions in India, has introduced the most advanced remotely operated vehicle (ROV) Mikros. The ROV, Micros, is poised to offer superior underwater inspection solutions to process industries, petrochemical refineries, and desalination plants. The partnership between companies to distribute remotely operated vehicles is boosting the global underwater drone market. For instance, September 15, 2017, UAV Propulsion Tech has signed a distributor agreement with Coral Partners to market their family of underwater marine remotely operated vehicles (ROV's). Nova Ray ROV's unique hydrodynamic features effectively use water flow. Nova Ray is customizable and available in four standard configurations based on depth of use (305m, 550m, 1500m, 4000m).

4.2.1 Market Size & Forecast

TABLE 2 REMOTELY OPERATED VEHICLE, UNDERWATER DRONE MARKET SHARE, BY REGION, 2016 – 2025 (\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	666	741	824	916	1,019	1,135	1,263	1,407	1,567	1,746	11.3%
Europe	487	541	602	669	744	828	921	1,025	1,142	1,272	11.3%
Asia-Pacific	395	442	494	552	618	691	774	867	971	1,088	12.0%
Rest of the World	146	162	180	199	221	246	273	303	336	373	11.0%
Total	1,694	1,885	2,099	2,337	2,602	2,899	3,230	3,601	4,015	4,478	11.4%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

4.3 Autonomous Underwater Vehicle

AUV Underwater Drone market segment is estimated to be \$842 million in 2018 and expected to garner \$ 2,624 million by 2025. North America is expected to be largest market for AUV Underwater Drone and is anticipated to grow at a CAGR of 16.7% from 2019-2025. AUV segment is projected to grow at a CAGR of 17.5% from 2019 to 2025.

Autonomous Underwater Vehicle (AUV) is used for underwater survey missions such as detecting and mapping submerged wrecks, rocks, and obstructions that can be a hazard to navigation for commercial and recreational vessels. AUVs carry equipment like video cameras, lights, robotic arms to grab things. AUVs are used as survey platforms to map the seafloor or characterize the physical, chemical, or biological properties of the water. AUV conducts its survey mission without operator intervention when a mission is complete, and the AUV will return to a pre-programmed location where the data can be downloaded and processed.

The launch of new Autonomous Underwater Vehicle (AUV) by the companies is boosting the global underwater drone market. For instance, on February 5, 2019, Teledyne, have newly designed SeaRaptor AUV incorporates a broad range of Teledyne content including acoustic modems, ascent and descent weight releases, a black box pinger locator, sub-bottom profiler (Teledyne Benthos), obstacle avoidance multi-beam sonar (Teledyne RESON), multi-beam echosounders, Doppler Velocity Log (DVL), Temperature, and Depth sensor (CTD) (Teledyne RD Instruments), Current, and onboard processing software (Teledyne Caris). The facility expansion by the company to develop, sale and post-sale servicing of AUVs is boosting the global underwater drone market. For instance, on February 1, 2019, Kawasaki Heavy Industries, Ltd. have expanded Kawasaki Subsea (UK) Limited, its new subsidiary in Aberdeen, Scotland, UK. Kawasaki Subsea (UK) Limited, its new subsidiary specializing in the production, sale, and post-sale servicing of AUVs, with a focus on the growing demand for pipeline maintenance in offshore oil and gas fields, Kawasaki has been developing an AUV equipped with a robot arm for performing subsea pipeline inspections.

4.3.1 Market Size & Forecast

TABLE 3 AUTONOMOUS UNDERWATER VEHICLE, UNDERWATER DRONE MARKET SHARE, BY REGION, 2016 – 2025 (\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	257	302	355	416	488	570	666	777	905	1,053	16.7%
Europe	168	200	237	280	331	390	458	538	631	738	17.5%
Asia-Pacific	123	150	181	219	263	316	377	450	535	635	19.4%
Rest of the World	50.2	58.9	68.9	80.5	93.8	109	127	147	171	197	16.1%
Total	598	710	842	996	1,175	1,384	1,628	1,911	2,241	2,624	17.5%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

5 Application Overview

5.1 Introduction

Global Underwater Drone market is segmented on the basis of Application by Commercial Exploration, Defense, Scientific Research, and Others. Commercial Exploration segment dominates the market share with the fastest growing segment of the global Underwater Drone market.

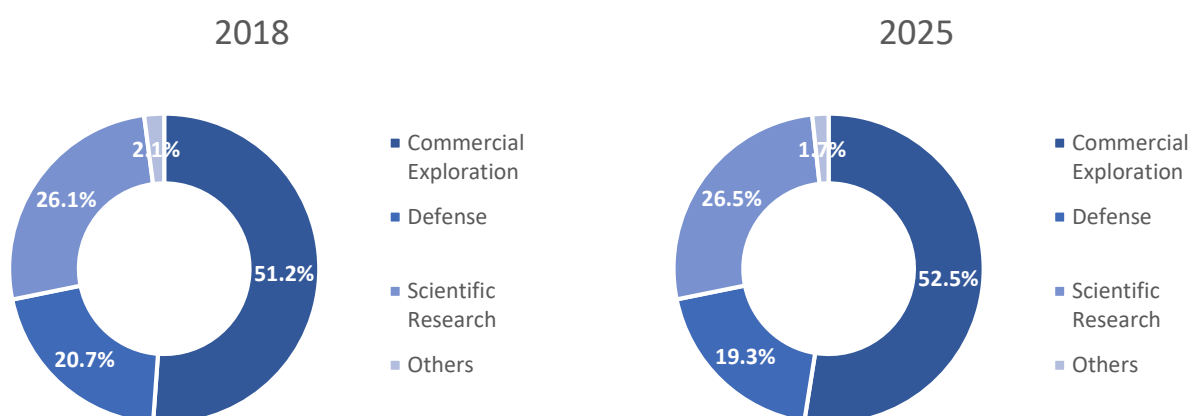
5.1.1 Market Size & Forecast

TABLE 4 GLOBAL UNDERWATER DRONE MARKET SHARE, BY APPLICATION, 2016 – 2025
(\$MILLION)

Application	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
Commercial Exploration	1,163	1,323	1,505	1,711	1,947	2,216	2,523	2,873	3,273	3,730	13.9%
Defense	483	542	608	682	766	860	966	1,085	1,220	1,372	12.3%
Scientific Research	596	676	768	872	990	1,125	1,279	1,454	1,653	1,880	13.7%
Others	49.6	55.1	61.2	67.8	75.0	82.8	91.2	100	110	120	10.0%
Total	2,292	2,596	2,942	3,333	3,778	4,284	4,858	5,513	6,256	7,102	13.4%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

FIGURE 3 GLOBAL UNDERWATER DRONE MARKET SHARE BY APPLICATION, 2018 VS 2025 (%)



Source: ERC Analysis, Government Publications, and Secondary Research

5.2 Commercial Exploration

Commercial Exploration Underwater Drone market segment is estimated to be \$ 1,505 million in 2018 and expected to garner \$ 3,730 million by 2025. North America is expected to be largest market for Commercial Exploration Underwater Drone and is anticipated to grow at a CAGR of 13.7% from 2019-2025. Commercial Exploration segment is projected to grow at a CAGR of 13.9% from 2019 to 2025.

Underwater drones are used for Commercial Exploration such as oil and renewables industries, and Infrastructure Surveys. Underwater ROVs became more widely used for the offshore oil and gas industry. Underwater drones operate at greater depths and carry much higher-resolution cameras, advanced sonar, and more capable manipulator arms. The inspection and maintenance of utilities are critical, and to maintain service and avoid hazards to navigation. ROV or AUV are used to inspect the deepest structures and bridges. Underwater drones can supply better methods for acquiring and recording data that improve quality, cost-effectiveness, and safety of traditional inspections.

The increasing agreement between companies for purchasing underwater drones for commercial exploration is boosting the market growth. For instance, on March 26, 2018, ECA Group has signed a cooperation agreement with Petrus to provide subsea robotics services to the oil and gas industry. Under the agreement, the companies will work together to offer inspection and survey services using the A18D AUV autonomous underwater vehicle (AUV). ECA Group's A18D AUV dive to a depth of 3000m and work autonomously for 24 hours. The AUV can carry out inspection and 3D mapping missions. Moreover, on July 27, 2017, Saudi Aramco has purchased Gavia Offshore Surveyor AUV system from Teledyne. The Gavia AUV system is expected to enhance Saudi Aramco's agility in meeting high demands for hydrographic survey services through the company's in-house survey operations. The Gavia system will be utilized predominantly for debris survey and pipeline inspection in the Arabian Gulf, with a potential service extension for other types of surveys in water depths up to 1000m in the Red Sea region. The Gavia AUV system purchased by Saudi Aramco is equipped with an Interferometric side scan sonar, camera, and utilizes a high accuracy Doppler Velocity Log aided Inertial Navigation System.

5.2.1 Market Size & Forecast

TABLE 5 COMMERCIAL EXPLORATION, UNDERWATER DRONE MARKET SHARE, BY REGION, 2016 – 2025 (\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	447	507	576	654	742	844	959	1,090	1,240	1,410	13.7%
Europe	336	382	434	493	560	636	724	823	937	1,066	13.7%
Asia-Pacific	278	319	365	418	479	548	629	721	827	948	14.6%
Rest of the World	102	115	130	147	166	188	212	239	270	305	12.9%
Total	1,163	1,323	1,505	1,711	1,947	2,216	2,523	2,873	3,273	3,730	13.9%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

5.3 Defense

Defense Underwater Drone market segment is estimated to be \$608 million in 2018 and expected to garner \$ 1,372 million by 2025. North America is expected to be largest market for Defense Underwater Drone and is anticipated to grow at a CAGR of 11.9% from 2019-2025. Defense segment is projected to grow at a CAGR of 12.3% from 2019 to 2025.

ROVs are used mainly for observation and inspection for the military sectors because they can only withstand shallow deployment (around 1,000 feet or 300 meters) and have the limited horsepower. The AUV provides mapping, imaging, underwater operational awareness and payload delivery for a variety of commercial and military missions. The increasing investment by government for the research activities of underwater drones is boosting the market. For instance, in 2017, the Department of Defense spend at least \$350 million on underwater drone procurement and research. The increasing contracts by the government for underwater drones to the companies have shown significant growth to the market. For instance, August 29, 2019, Teledyne Instruments, Inc., has received an indefinite delivery/indefinite quantity (IDIQ) contract by the U.S. Naval Undersea Warfare Center to supply autonomous underwater vehicles (AUVs) and related monitoring and communications acoustic systems. The contract is approximately \$5.4 million. The award has a maximum value of \$22.2 million through December 2024, if all contract options are exercised.

5.3.1 Market Size & Forecast

TABLE 6 DEFENSE, UNDERWATER DRONE MARKET SHARE, BY REGION, 2016 – 2025 (\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	207	231	258	288	322	360	403	451	505	565	11.9%
Europe	132	148	166	186	209	234	263	295	331	372	12.2%
Asia-Pacific	105	119	135	153	174	197	224	255	289	329	13.6%
Rest of the World	39.6	44.1	49.3	54.9	61.3	68.3	76.2	85.1	94.9	106	11.6%
Total	483	542	608	682	766	860	966	1,085	1,220	1,372	12.3%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

5.4 Scientific Research

Scientific Research Underwater Drone market segment is estimated to be \$768 million in 2018 and expected to garner \$ 1,880 million by 2025. North America is expected to be largest market for Scientific Research Underwater Drone and is anticipated to grow at a CAGR of 13.4% from 2019-2025. Scientific Research segment is projected to grow at a CAGR of 13.7% from 2019 to 2025.

Underwater drones are helping scientists to explore and understand the undersea world and protecting it. The Underwater drones also provide water sampler to retrieve samples of water for research and tests in oceans, lakes, and rivers. The researchers are helping save the underwater environment, particularly in areas that are inaccessible and too deep for divers. From cleaning the oil spill off the Gulf of Mexico to monitoring Great White Sharks near Guadalupe, underwater drones are helping to protect our oceans. The Water Sampler allows you to retrieve samples of water for research and tests in oceans, lakes, and rivers. The introduction of new underwater drone for scientific research activity is boosting the market growth. For instance, on December 19, 2017, Aquabotix has introduced the new Integra AUV/ROV (autonomous underwater vehicle/remotely operated vehicle) for research, environmental assessment. The second-generation hybrid underwater vehicle is powered via a lithium-ion battery, making it possible for users. It can be configured with multiple sensors and maneuvered with an intuitive platform that's accessible from Web-enabled devices.

5.4.1 Market Size & Forecast

TABLE 7 SCIENTIFIC RESEARCH, UNDERWATER DRONE MARKET SHARE, BY REGION, 2016 – 2025 (\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	244	276	313	354	402	455	517	586	665	755	13.4%
Europe	175	199	225	255	289	328	372	423	480	545	13.5%
Asia-Pacific	125	143	164	188	215	247	283	325	372	427	14.6%
Rest of the World	51.8	58.4	65.9	74.3	83.8	94.6	107	120	136	153	12.8%
Total	596	676	768	872	990	1,125	1,279	1,454	1,653	1,880	13.7%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

5.5 Others

Others Underwater Drone market segment is estimated to be \$61.2 million in 2018 and expected to garner \$120 million by 2025. North America is expected to be largest market for Others Underwater Drone and is anticipated to grow at a CAGR of 11.0% from 2019-2025. Others segment is projected to grow at a CAGR of 10.0% from 2019 to 2025.

The other application of underwater drone includes Search & Rescue. The law Enforcement uses underwater drone for Search & Rescue mission. It is used to identify the missing person or item in the deep sea. The underwater drone for Search & Rescue mission are equipped for a quick 30-second deployment time, and its camera allows for clear visibility underwater and up to 1000 feet (300 meters) in depth. The introduction of new underwater drone for search and rescue operation is boosting the market growth. For instance, on October 29, 2018, Notilo Plus, a company developing intelligent, autonomous underwater systems, has introduced iBubble, the world's first fully wireless, autonomous and intelligent underwater drone. iBubble can accompany divers for up to an hour and up to 197 feet depth, acting as their personal cameraman, enabling superior underwater imaging. Bubble can be used for search and rescue operations.

5.5.1 Market Size & Forecast

TABLE 8 OTHERS, UNDERWATER DRONE MARKET SHARE, BY REGION, 2016 – 2025 (\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	26.3	29.4	32.9	36.7	40.8	45.4	50.4	55.9	61.9	68.4	11.0%
Europe	10.9	12.1	13.5	15.1	16.8	18.6	20.6	22.8	25.1	27.7	10.6%
Asia-Pacific	9.77	10.66	11.59	12.56	13.56	14.60	15.63	16.66	17.64	18.54	6.7%
Rest of the World	2.65	2.91	3.20	3.50	3.83	4.17	4.53	4.90	5.29	5.68	8.4%
Total	49.6	55.1	61.2	67.8	75.0	82.8	91.2	100.3	110.0	120.3	10.0%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

6 Component Overview

6.1 Introduction

Global Underwater Drone market is segmented on the basis of Component by Sensors, Synthetic Aperture Sonar, Inertial Navigation Systems, Imaging Systems, Lighting Systems, and Others. Imaging Systems segment dominates the market share. However, Sensors is the fastest growing segment of the global Underwater Drone market.

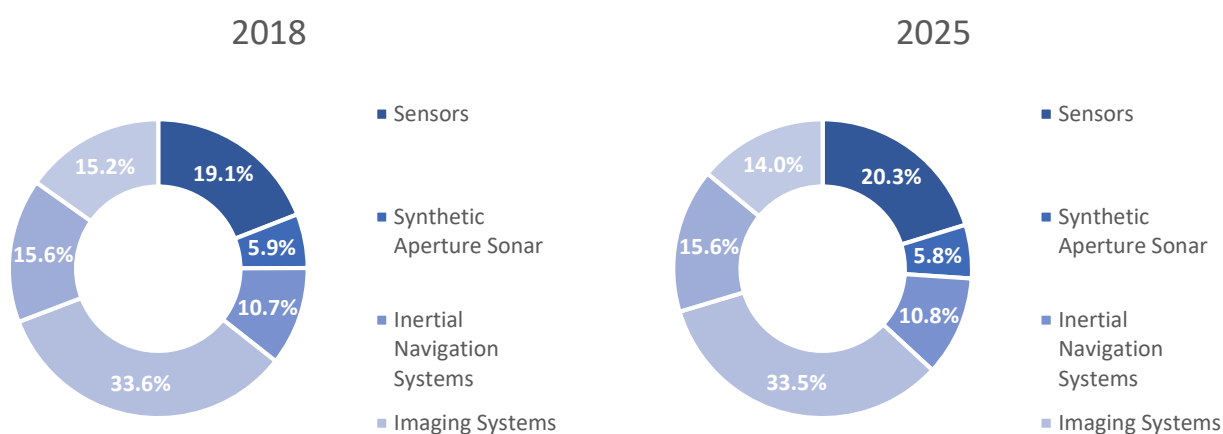
6.1.1 Market Size & Forecast

TABLE 9 GLOBAL UNDERWATER DRONE MARKET SHARE, BY COMPONENT, 2016 – 2025
(\$MILLION)

Component	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
Sensors	430	491	561	641	733	838	959	1,098	1,256	1,439	14.4%
Synthetic Aperture Sonar	135	152	172	195	221	250	283	320	363	411	13.3%
Inertial Navigation Systems	244	277	314	357	405	460	523	594	676	768	13.6%
Imaging Systems	772	873	988	1,119	1,267	1,436	1,628	1,846	2,095	2,378	13.4%
Lighting Systems	357	405	459	520	590	669	759	861	978	1,110	13.5%
Others	355	398	447	501	562	631	707	793	888	995	12.1%
Total	2,292	2,596	2,942	3,333	3,778	4,284	4,858	5,513	6,256	7,102	13.4%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

FIGURE 4 GLOBAL UNDERWATER DRONE MARKET SHARE BY COMPONENT, 2018 VS 2025 (%)



Source: ERC Analysis, Government Publications, and Secondary Research

6.2 Sensors

Sensors Underwater Drone market segment is estimated to be \$561 million in 2018 and expected to garner \$ 1,439 million by 2025. North America is expected to be largest market for Sensors Underwater Drone and is anticipated to grow at a CAGR of 14.1% from 2019-2025. Sensors segment is projected to grow at a CAGR of 14.4% from 2019 to 2025.

The underwater drone sensor is a small component designed to gather essential data. The sensor system is based on 5-DOF Inertial Measurement Unit sensor, MPX pressure sensor, and DS temperature sensor. These types of sensors integrated with signal processing hardware in one compact device. The underwater drone sensor can reduce space and weight in order to put other components. The sensors are used to explore the depths of the ocean, measure and help in the rescue operation and to maintain the safety of soldiers and sailors. It is also used in the ordinary military mission to map and detect or determine whether the seabed has mines. It is also can be used to detect an unidentified object. The main idea to use the sensors is to reduce the estimation of error, lower in cost, small space and improving the accuracy of range measurements.

6.2.1 Market Size & Forecast

TABLE 10 SENSORS, UNDERWATER DRONE MARKET SHARE, BY REGION, 2016 – 2025 (\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	177	202	230	262	299	341	389	444	507	579	14.1%
Europe	122	140	159	182	208	238	272	310	355	406	14.3%
Asia-Pacific	93.9	108	125	144	166	191	220	254	294	339	15.4%
Rest of the World	36.3	41.2	46.8	53.2	60.4	68.6	77.9	88.6	101	114	13.6%
Total	430	491	561	641	733	838	959	1,098	1,256	1,439	14.4%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

6.3 Synthetic Aperture Sonar

Synthetic Aperture Sonar Underwater Drone market segment is estimated to be \$172 million in 2018 and expected to garner \$411 million by 2025. North America is expected to be largest market for Synthetic Aperture Sonar Underwater Drone and is anticipated to grow at a CAGR of 13.0% from 2019-2025. Synthetic Aperture Sonar segment is projected to grow at a CAGR of 13.3% from 2019 to 2025.

Synthetic aperture sonar (SAS) is a form of sonar in which sophisticated post-processing of sonar data is used to produce very narrow effective beam. SAS is a good alternative for large AUVs, but require additional logistics, larger vessels, and crews to operate. SAS can only be used by moving instruments over relatively immobile targets, and it is used for wide applications in remote sensing and mapping. The launch of new Synthetic aperture sonar for underwater drones is boosting the market. For instance, March 6, 2019, Kraken Robotics Inc. has introduced new synthetic aperture sonar (SAS) to provide ultra-high definition seabed mapping with superior area coverage rates. the AquaPix MINSAS-240 SAS has been designed to advance the state-of-the-art in seabed mapping for both high-speed towed systems and large diameter unmanned underwater vehicles (LDUUVs)

6.3.1 Market Size & Forecast

TABLE 11 SYNTHETIC APERTURE SONAR, UNDERWATER DRONE MARKET SHARE, BY REGION, 2016 – 2025 (\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	53.5	60.4	68.2	76.9	86.8	98.1	111	125	142	160	13.0%
Europe	39.3	44.4	50.2	56.8	64.2	72.6	82.1	92.9	105	119	13.1%
Asia-Pacific	30.2	34.4	39.2	44.7	51.0	58.2	66.4	75.9	86.6	99.0	14.2%
Rest of the World	11.7	13.1	14.7	16.6	18.6	20.9	23.5	26.4	29.7	33.4	12.4%
Total	135	152	172	195	221	250	283	320	363	411	13.3%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

6.4 Inertial Navigation Systems (INS)

INS Underwater Drone market segment is estimated to be \$314 million in 2018 and expected to garner \$768 million by 2025. North America is expected to be largest market for INS Underwater Drone and is anticipated to grow at a CAGR of 13.4% from 2019-2025. INS segment is projected to grow at a CAGR of 13.6% from 2019 to 2025.

Inertial Navigation Systems (INS) helps in finding missing objects underwater, search operations, monitoring underwater mines, and surveying of offshore assets. Inertial Navigation Systems are required for localization, positioning, path tracking, guidance, and control. The navigational sensor outfit for an underwater vehicle consist of standard components such as a compass, pressure depth sensor, and INS. The incorporation of Inertial Navigation Systems in underwater drones to make it efficient for various application. For instance, on April 10, 2019, The National Oceanography Centre (NOC), has selected high-performance hybrid navigation technology from Sonardyne International Ltd. for the next generation of its Autosub autonomous underwater vehicle (AUV), targeted for under-ice operations. The NOC has incorporated Sonardyne's highest performing SPRINT-Nav inertial navigation system (INS) into the new 2,000 meters' depth-rated Autosub, which is being developed explicitly for carrying high-performance sensors on the most demanding research missions under polar ice, hence its name Autosub2KUI, or A2KUI.

6.4.1 Market Size & Forecast

TABLE 12 INERTIAL NAVIGATION SYSTEMS, UNDERWATER DRONE MARKET SHARE, BY REGION, 2016 – 2025 (\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	99.3	112	127	144	163	185	210	238	270	306	13.4%
Europe	70.2	79.5	90.2	102	116	132	149	170	193	219	13.5%
Asia-Pacific	53.9	61.6	70.5	81	92	106	121	139	159	182	14.5%
Rest of the World	20.8	23.5	26.5	29.9	33.7	38.0	42.8	48.3	54.5	61.5	12.8%
Total	244	277	314	357	405	460	523	594	676	768	13.6%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

6.5 Imaging Systems

Imaging Systems Underwater Drone market segment is estimated to be \$988 million in 2018 and expected to garner \$ 2,378 million by 2025. North America is expected to be largest market for Imaging Systems Underwater Drone and is anticipated to grow at a CAGR of 13.1% from 2019-2025. Imaging Systems segment is projected to grow at a CAGR of 13.4% from 2019 to 2025.

The imaging system includes cameras and video screens. Underwater drones with cameras that enable to see underwater view without having to get into the water. The cameras of underwater drones perform frequent inspections of stormwater/drain pipes, water tanks, sewer pipes for infrastructure maintenance, and to ensure the piping systems are functioning properly and are free of damage. Underwater drone cameras can assist in the installation and construction of offshore wind farms by enabling the monitoring of divers as they work and performing inspections of newly installed equipment. The underwater robotic cameras record events automatically in a specified area. The underwater video system is beneficial for activities such as monitoring aquatic wildlife, conducting academic research or maintaining security at an undersea drilling platform or port.

6.5.1 Market Size & Forecast

TABLE 13 IMAGING SYSTEMS, UNDERWATER DRONE MARKET SHARE, BY REGION, 2016 – 2025
(\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	314	354	400	451	510	577	652	738	835	946	13.1%
Europe	222	251	284	321	363	411	465	527	598	678	13.3%
Asia-Pacific	170	194	222	253	289	330	377	431	493	564	14.3%
Rest of the World	65.8	74.0	83.2	93.6	105	118	133	150	169	190	12.6%
Total	772	873	988	1,119	1,267	1,436	1,628	1,846	2,095	2,378	13.4%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

6.6 Lighting Systems

Lighting Systems Underwater Drone market segment is estimated to be \$459 million in 2018 and expected to garner \$ 1,110 million by 2025. North America is expected to be largest market for Lighting Systems Underwater Drone and is anticipated to grow at a CAGR of 13.2% from 2019-2025. Lighting Systems segment is projected to grow at a CAGR of 13.5% from 2019 to 2025.

The lighting system is used for clear visibility and identifies thing surrounded to the underwater drones. The lighting system is used for inspections of stormwater/drain pipes, water tanks, sewer pipes for infrastructure maintenance, and to ensure the piping systems are functioning properly and are free of damage. It minimizes the blurred content in the video or photos.

6.6.1 Market Size & Forecast

TABLE 14 LIGHTING SYSTEMS, UNDERWATER DRONE MARKET SHARE, BY REGION, 2016 – 2025
(\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	145	163	185	209	236	267	303	343	388	440	13.2%
Europe	103	117	132	150	169	192	218	247	280	317	13.4%
Asia-Pacific	79.1	90.3	103	118	135	154	176	202	231	264	14.4%
Rest of the World	30.5	34.4	38.8	43.6	49.1	55.4	62.4	70.2	79.1	89.1	12.6%
Total	357	405	459	520	590	669	759	861	978	1,110	13.5%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

6.7 Others

Others Underwater Drone market segment is estimated to be \$447 million in 2018 and expected to garner \$995 million by 2025. North America is expected to be largest market for Others Underwater Drone and is anticipated to grow at a CAGR of 11.70% from 2019-2025. Others segment is projected to grow at a CAGR of 12.10% from 2019 to 2025.

The other components of underwater drones include Steering and Positioning, Energy and Propulsion, and Automation Systems. Steering and Positioning give direction to the drone by the controller. The underwater drones use rechargeable lithium batteries, some of the larger AUVs use aluminum-based semi-fuel cells that are higher-maintenance which make them work for days, even weeks, without human intervention. Researchers are trying to make AUVs more independent by developing machine intelligence solutions that enable AUVs to navigate complex situations and solve problems by themselves.

6.7.1 Market Size & Forecast

TABLE 15 OTHERS, UNDERWATER DRONE MARKET SHARE, BY REGION, 2016 – 2025 (\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	135	151	169	189	212	236	264	295	330	368	11.7%
Europe	97.8	110	123	138	154	173	193	216	242	270	11.9%
Asia-Pacific	90.6	103	116	132	149	168	190	215	244	275	13.1%
Rest of the World	31.0	34.6	38.6	43.0	47.9	53.3	59.3	66.0	73.4	81.5	11.2%
Total	355	398	447	501	562	631	707	793	888	995	12.1%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

7 Propulsion System Overview

7.1 Introduction

Global Underwater Drone market is segmented on the basis of Propulsion System by Electric System, Mechanical System, and Hybrid Systems. Electric System segment dominates the market share with the fastest growing segment of the global Underwater Drone market.

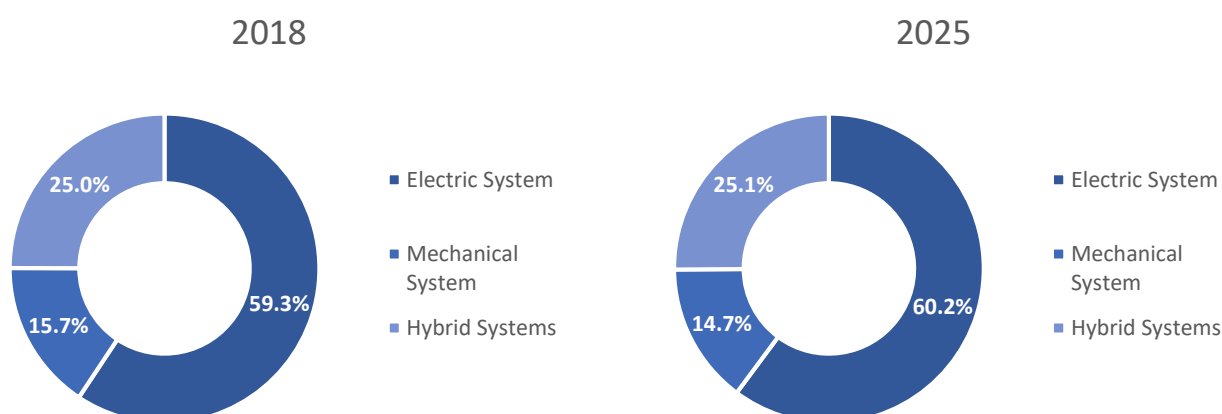
7.1.1 Market Size & Forecast

TABLE 16 GLOBAL UNDERWATER DRONE MARKET SHARE, BY PROPULSION SYSTEM, 2016 – 2025 (\$MILLION)

Propulsion System	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
Electric System	1,354	1,537	1,745	1,982	2,250	2,557	2,905	3,303	3,756	4,273	13.7%
Mechanical System	367	412	462	519	583	655	736	827	930	1,046	12.4%
Hybrid Systems	571	647	734	833	945	1,072	1,217	1,382	1,570	1,784	13.5%
Total	2,292	2,596	2,942	3,333	3,778	4,284	4,858	5,513	6,256	7,102	13.4%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

FIGURE 5 GLOBAL UNDERWATER DRONE MARKET SHARE BY PROPULSION SYSTEM, 2018 VS 2025 (%)



Source: ERC Analysis, Government Publications, and Secondary Research

7.2 Electric System

Electric System Underwater Drone market segment is estimated to be \$ 1,745 million in 2018 and expected to garner \$ 4,273 million by 2025. North America is expected to be largest market for Electric System Underwater Drone and is anticipated to grow at a CAGR of 13.3% from 2019-2025. Electric System segment is projected to grow at a CAGR of 13.7% from 2019 to 2025.

Electric system use battery to supply power. Electric propulsion offers clean operation with excellent reliability and simple maintenance. Electric propulsion has outstanding low noise and low vibration characteristics. The launch of new electric propulsion underwater drones is boosting the global underwater drone market. For instance, on September 7, 2019, SMD a global supplier of subsea equipment, have developed a new work-class ROV (Remotely Operated Vehicle), the Quantum EV, with electric drive technology and a modular design. The Quantum EV ROV is intended to be an environmentally responsible solution. Moreover, on October 29, 2018, Notilo Plus, a company developing intelligent, autonomous underwater systems, has introduced iBubble, the world's first fully wireless, autonomous and intelligent underwater drone. iBubble can accompany divers for up to an hour and up to 197 feet depth, acting as their personal cameraman, enabling superior underwater imaging.

7.2.1 Market Size & Forecast

TABLE 17 ELECTRIC SYSTEM, UNDERWATER DRONE MARKET SHARE, BY REGION, 2016 – 2025
(\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	557	631	714	809	916	1,038	1,176	1,333	1,512	1,715	13.3%
Europe	383	434	492	558	633	719	816	927	1,053	1,197	13.5%
Asia-Pacific	298	341	390	447	512	587	673	772	886	1,016	14.7%
Rest of the World	116	131	148	167	189	213	240	271	306	345	12.8%
Total	1,354	1,537	1,745	1,982	2,250	2,557	2,905	3,303	3,756	4,273	13.7%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

7.3 Mechanical System

Mechanical System Underwater Drone market segment is estimated to be \$462 million in 2018 and expected to garner \$ 1,046 million by 2025. North America is expected to be largest market for Mechanical System Underwater Drone and is anticipated to grow at a CAGR of 12.3% from 2019-2025. Mechanical System segment is projected to grow at a CAGR of 12.4% from 2019 to 2025.

The rise in popularity of mechanical underwater drones is highly influenced by the lack of complications associated with electrical systems and easy availability of modules and associated parts for smooth operations. The mechanical systems provide flexibility of movement, with drones able to tilt up to 45 degrees upward and downward, making it possible for it to take shots from multiple angles while navigating complex undersea topographies like the coral reefs. The companies recognize the positive market traction gained by mechanical underwater drones as means to increase financial product value and extend consumer base across global locations. For instance, on June 24, 2019, Shanghai Youcan Robot, a high-tech firm focused on the design and production of auto-piloting underwater drones and applications, has unveiled its latest generation of civilian unmanned underwater drone named BW Space Pro. The product has made a shift from the old model's three-rotary-blade power system to one composed of four rotary blades, a change that allows for more flexible vertical movement in the water.

7.3.1 Market Size & Forecast

TABLE 18 MECHANICAL SYSTEM, UNDERWATER DRONE MARKET SHARE, BY REGION, 2016 – 2025 (\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	142	159	178	200	224	251	282	317	356	400	12.3%
Europe	107	119	134	150	169	190	213	239	269	302	12.3%
Asia-Pacific	87.5	98.6	111	125	142	160	180	204	230	260	12.9%
Rest of the World	31.1	34.7	38.8	43.3	48.3	53.9	60.1	67.1	74.9	83.6	11.6%
Total	367	412	462	519	583	655	736	827	930	1,046	12.4%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

7.4 Hybrid Systems

Hybrid Systems Underwater Drone market segment is estimated to be \$734 million in 2018 and expected to garner \$ 1,784 million by 2025. North America is expected to be largest market for Hybrid Systems Underwater Drone and is anticipated to grow at a CAGR of 13.2% from 2019-2025. Hybrid Systems segment is projected to grow at a CAGR of 13.5% from 2019 to 2025.

The Hybrid underwater drones can function both as an autonomous underwater vehicle (AUV) and as a remotely-operated vehicle (ROV). The design cost-effectiveness and ability to conduct multiple underwater missions without the need to deploy separate ROVs and AUVs are the primary product features which would increase sales volume across numerous global locations. The Hybrid underwater drones search wide areas using AUV mode (untethered) while conducting detailed inspections using ROV mode (tethered). The companies are thus focused on launch of new, innovative products as means to increase product sales and strengthen global market position. For instance, on January 03, 2018, Australia-based UUV Aquabotix Limited announced the launch of its second-generation hybrid, underwater drone, Integra. The Integra is equipped with configurable sensor arrays and can be deployed by a single user. Aquabotix officials say the AUV is ideal for scientific research, inspections or law-enforcement/defense operations.

7.4.1 Market Size & Forecast

TABLE 19 HYBRID SYSTEMS, UNDERWATER DRONE MARKET SHARE, BY REGION, 2016 – 2025
(\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	224	254	287	325	367	416	470	533	604	684	13.2%
Europe	165	187	212	240	272	309	350	397	450	511	13.4%
Asia-Pacific	133	152	174	199	227	260	298	341	390	447	14.5%
Rest of the World	48.5	54.7	61.6	69.4	78.2	88.1	99.3	112	126	142	12.7%
Total	571	647	734	833	945	1,072	1,217	1,382	1,570	1,784	13.5%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8 Regional Overview

8.1 Introduction

Global Underwater Drone market is segmented by region as North America, Europe, Asia-Pacific, and Rest of the World. North America is further segmented into the US, Canada, and Mexico. Europe region covered United Kingdom, France, Germany, Russia, and Rest of Europe. Asia-Pacific region consists of China, India, Japan, South Korea, and Rest of Asia-pacific. Rest of the World segment cover Middle East, Africa, and Rest of the World region.

8.1.1 Market Size & Forecast

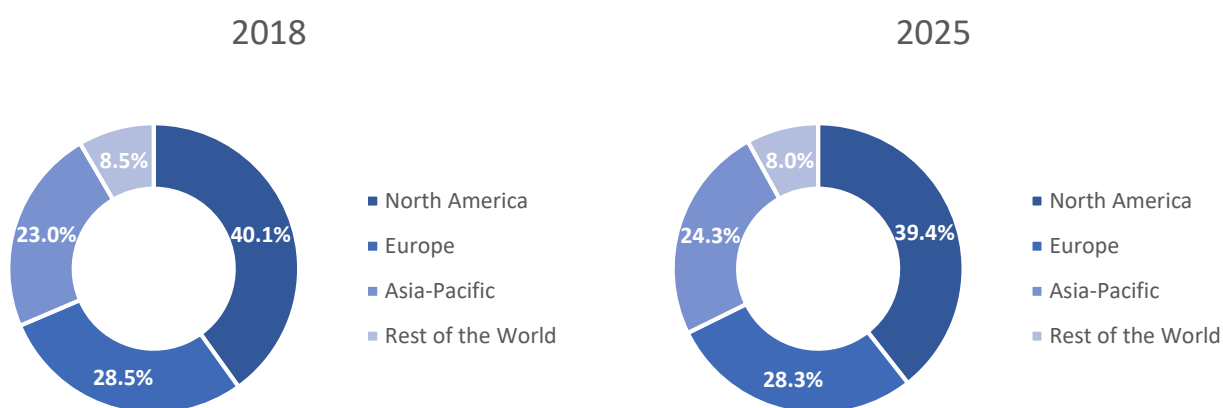
TABLE 20 GLOBAL UNDERWATER DRONE MARKET SHARE VALUE, 2016 – 2025 (\$MILLION)

Region	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
North America	923	1,043	1,179	1,333	1,507	1,705	1,929	2,183	2,472	2,799	13.2%
Europe	655	741	838	949	1,075	1,217	1,379	1,563	1,772	2,010	13.3%
Asia-Pacific	518	591	675	771	881	1,007	1,151	1,316	1,506	1,723	14.3%
Rest of the World	196	221	249	280	315	355	399	450	506	570	12.6%
Total	2,292	2,596	2,942	3,333	3,778	4,284	4,858	5,513	6,256	7,102	13.4%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

FIGURE 6

GLOBAL UNDERWATER DRONE MARKET SHARE BY REGION, 2018 VS 2025 (%)



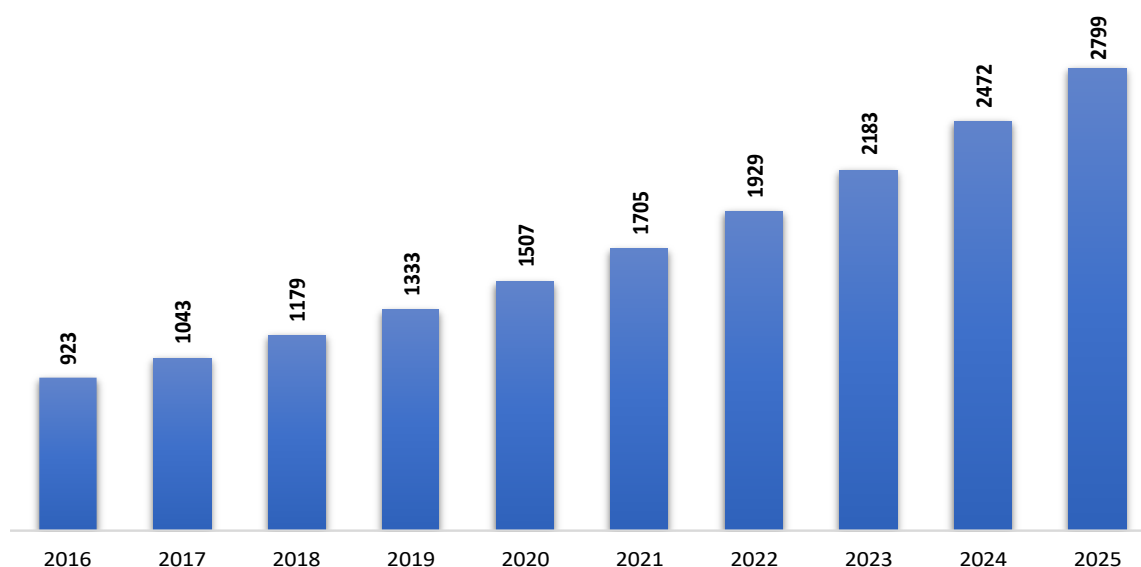
Source: ERC Analysis, Government Publications, and Secondary Research

8.2 North America

The continuous exploration activities of shale oil and attempts to discover additional offshore resources in deep depths of the oceans are the influential primary factors that would propel the growth prospects of the Underwater Drones market in North America during the forecast period. The rise in deployment of underwater vehicles for the detection of military threats to public safety & security would also improve product application in the region. The companies are thus focused on formation of strategic collaborations and partnerships in a bid to strengthen business position in the regional market and expand business activities. For instance, on August 19, 2019, Teledyne Technologies Incorporated announced that its subsidiary, Teledyne Instruments, Inc., has been awarded an indefinite-delivery/indefinite-quantity (IDIQ) contract by the U.S. Naval Undersea Warfare Center to supply autonomous underwater vehicles (AUVs) and related monitoring and communications acoustic systems. The initial 2019 value of the contract is approximately \$5.4 million. The successful deployment of underwater drones during emergencies by various military groups in the region have also improved the overall market growth rate. For instance, on November 19, 2017, the U.S. Navy deployed unmanned underwater vehicles (UUV) to join in the search for the Argentine navy's submarine, A.R.A. San Juan, in South Atlantic waters. The equipment consists of one Bluefin 12D (Deep) UUV and three Iver 580 UUVs, which are operated by the U.S. Navy's recently-established Unmanned Undersea Vehicle Squadron 1, based in Keyport, Washington.

8.2.1 Market Size & Forecast

TABLE 21 NORTH AMERICA, UNDERWATER DRONE MARKET, 2016 – 2025 (\$MILLION)



Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.2.2 North America Market Size & Forecast, By Country

TABLE 22 NORTH AMERICA UNDERWATER DRONE MARKET SHARE VALUE, BY COUNTRY, 2016 – 2025 (\$MILLION)

Country	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
U.S.	549	621	703	796	901	1,021	1,156	1,311	1,486	1,685	13.3%
Canada	263	296	334	377	426	481	543	614	694	784	13.0%
Mexico	112	126	142	160	180	203	229	259	292	330	12.9%
Total	923	1,043	1,179	1,333	1,507	1,705	1,929	2,183	2,472	2,799	13.2%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.2.3 North America Market Size & Forecast, By Type

TABLE 23 NORTH AMERICA UNDERWATER DRONE MARKET SHARE VALUE, BY TYPE, 2016 – 2025 (\$MILLION)

Type	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
ROV	666	741	824	916	1,019	1,135	1,263	1,407	1,567	1,746	11.3%
AUV	257	302	355	416	488	570	666	777	905	1,053	16.7%
Total	923	1,043	1,179	1,333	1,507	1,705	1,929	2,183	2,472	2,799	13.2%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.2.4 North America Market Size & Forecast, By Application

TABLE 24 NORTH AMERICA UNDERWATER DRONE MARKET SHARE VALUE, BY APPLICATION, 2016 – 2025 (\$MILLION)

Application	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
Commercial Exploration	447	507	576	654	742	844	959	1,090	1,240	1,410	13.7%
Defense	207	231	258	288	322	360	403	451	505	565	11.9%
Scientific Research	244	276	313	354	402	455	517	586	665	755	13.4%
Others	26.3	29.4	32.9	36.7	40.8	45.4	50.4	55.9	61.9	68.4	11.0%
Total	923	1,043	1,179	1,333	1,507	1,705	1,929	2,183	2,472	2,799	13.2%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.2.5 North America Market Size & Forecast, By Component

TABLE 25 NORTH AMERICA UNDERWATER DRONE MARKET SHARE VALUE, BY COMPONENT, 2016 – 2025 (\$MILLION)

Component	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
Sensors	177	202	230	262	299	341	389	444	507	579	14.1%
Synthetic Aperture Sonar	53.5	60.4	68.2	76.9	86.8	98.1	111	125	142	160	13.0%
Inertial Navigation Systems	99.3	112	127	144	163	185	210	238	270	306	13.4%
Imaging Systems	314	354	400	451	510	577	652	738	835	946	13.1%
Lighting Systems	145	163	185	209	236	267	303	343	388	440	13.2%
Others	135	151	169	189	212	236	264	295	330	368	11.7%
Total	923	1,043	1,179	1,333	1,507	1,705	1,929	2,183	2,472	2,799	13.2%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.2.6 North America Market Size & Forecast, By Propulsion System

TABLE 26 NORTH AMERICA UNDERWATER DRONE MARKET SHARE VALUE, BY PROPULSION SYSTEM, 2016 – 2025 (\$MILLION)

Propulsion System	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
Electric System	557	631	714	809	916	1,038	1,176	1,333	1,512	1,715	13.3%
Mechanical System	142	159	178	200	224	251	282	317	356	400	12.3%
Hybrid Systems	224	254	287	325	367	416	470	533	604	684	13.2%
Total	923	1,043	1,179	1,333	1,507	1,705	1,929	2,183	2,472	2,799	13.2%

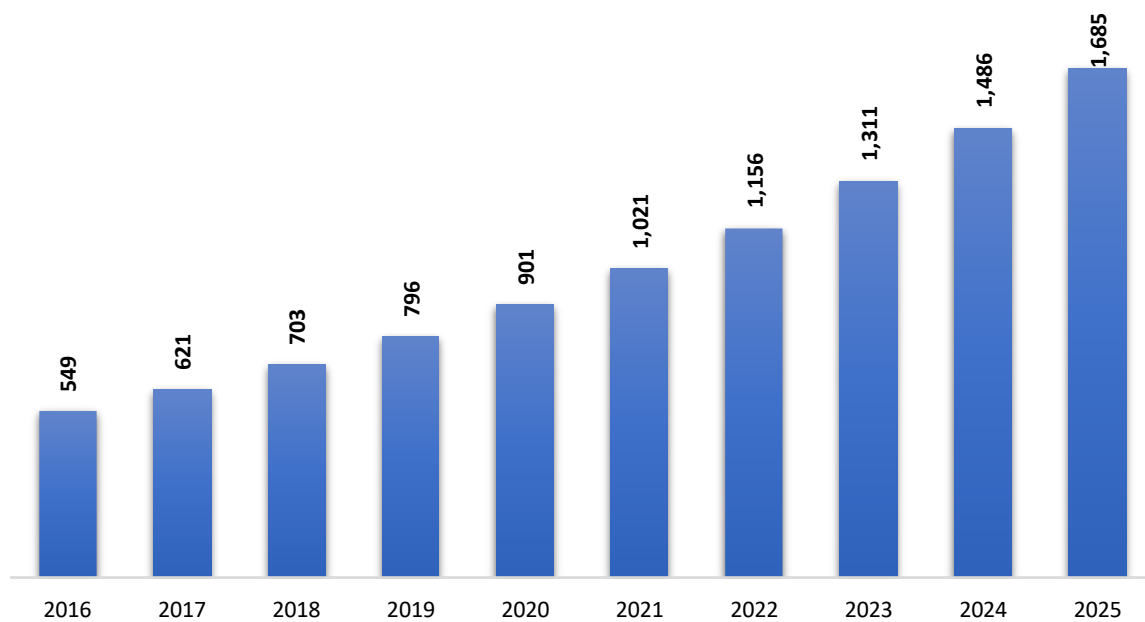
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.2.7 U.S.

U.S. Underwater Drone market was valued at \$703 million in 2018 and is expected to reach a value of \$ 1,685 million by the end of 2025, growing at a CAGR of 13.3% from 2019 to 2025.

8.2.7.1 Market Size and Forecast

FIGURE 7 U.S. UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



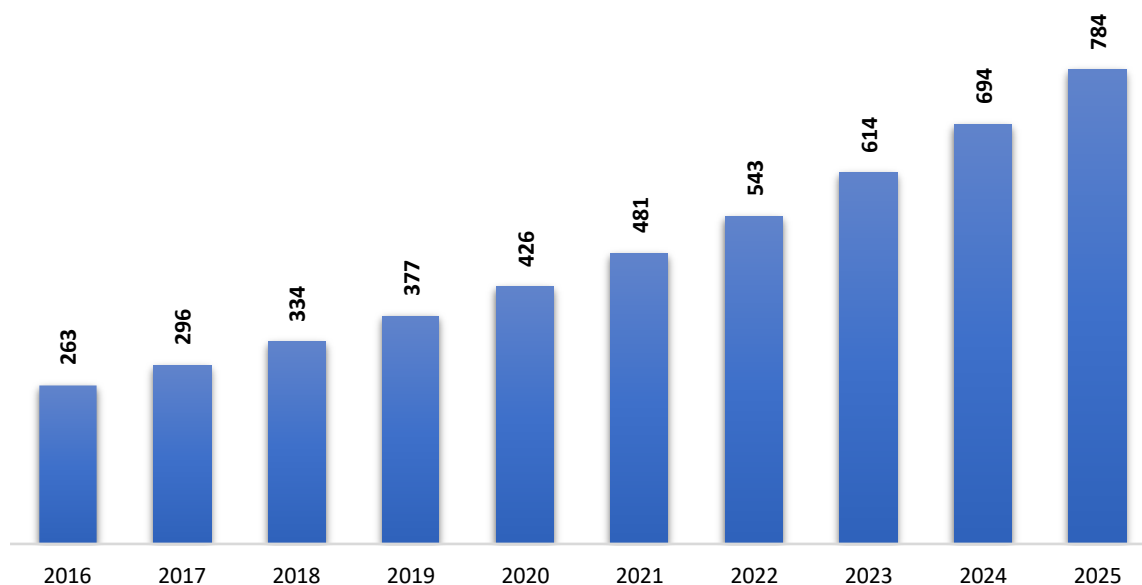
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.2.8 Canada

Canada Underwater Drone market was valued at \$334 million in 2018 and is expected to reach a value of \$784 million by the end of 2025, growing at a CAGR of 13.0% from 2019 to 2025.

8.2.8.1 Market Size and Forecast

FIGURE 8 CANADA UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



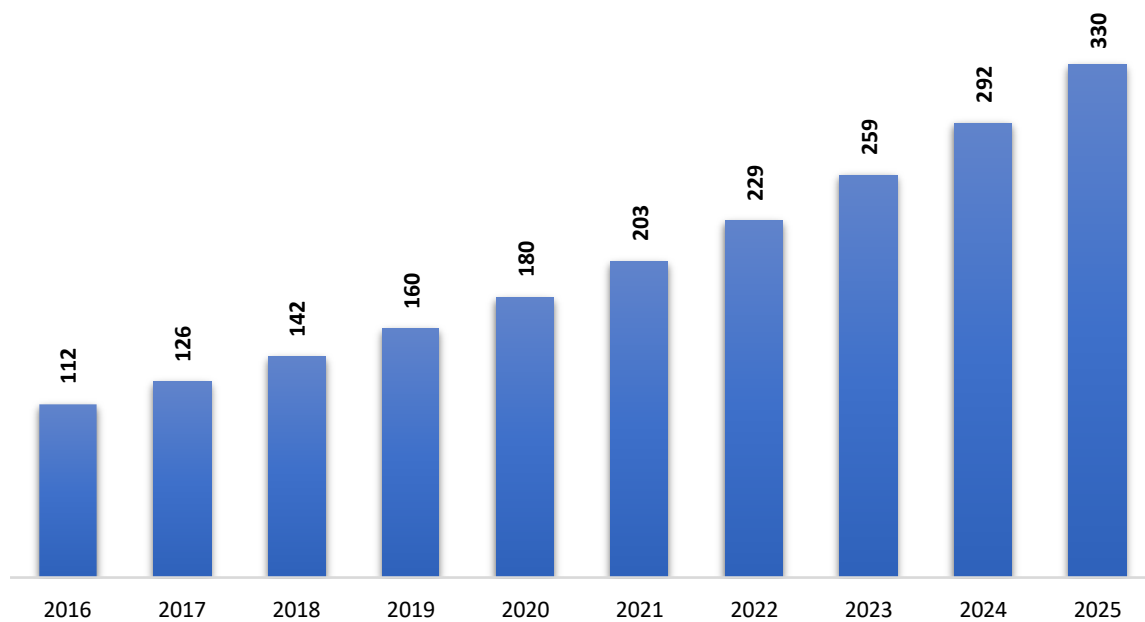
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.2.9 Mexico

Mexico Underwater Drone market was valued at \$142 million in 2018 and is expected to reach a value of \$330 million by the end of 2025, growing at a CAGR of 12.9% from 2019 to 2025.

8.2.9.1 Market Size and Forecast

FIGURE 9 MEXICO UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



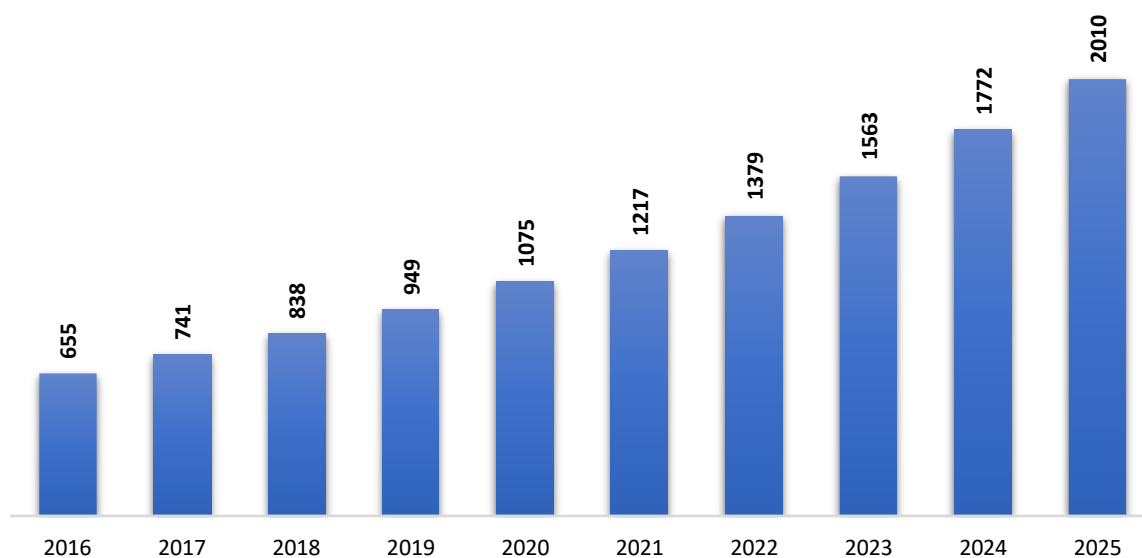
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.3 Europe

The rapid increase in deployment of underwater drones for surveillance, discovery, and exploration activities in Europe has led to a definite increase in product demand in the region, with scientific research into deep-water life sciences by various research institutes also improving the overall growth prospects of the European market. The companies recognize the positive gain in market traction as means to increase business activities and strengthen business position. For instance, on January 11, 2019, the Russian Navy announced plans to place more than 30 Poseidon strategic nuclear-capable underwater drones on combat duty. Further, on April 17, 2019, the British government launched a competition to test a large, unmanned submarine's ability to gather intelligence and perform other roles. The Ministry of Defence said it would provide USD 3.3 million for the winning bidder to design, refit an existing platform and trial the utility of extra-large unmanned underwater vehicles, otherwise known as XLUUVs. The three-year effort includes yearlong research, design and vehicle refit as the first stage, followed by a two-year program to test a sizeable unmanned submarine's ability to undertake a series of roles. Moreover, on June 26, 2017, Kraken Sonar Inc. announced that its wholly-owned subsidiary, Kraken Sonar Systems Inc. has formally taken delivery of ThunderFish Alpha, a next-generation Autonomous Underwater Vehicle (AUV) designed for deep-sea military, commercial and scientific applications. The ThunderFish Alpha AUV was developed by Germany's Fraunhofer Institute and will be used as a technology demonstrator platform to support ongoing development of Kraken's underwater sensor and robotics programs.

8.3.1 Market Size & Forecast

TABLE 27 EUROPE, UNDERWATER DRONE MARKET, 2016 – 2025 (\$MILLION)



Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.3.2 Europe Market Size & Forecast, By Country

TABLE 28 EUROPE UNDERWATER DRONE MARKET SHARE VALUE, BY COUNTRY, 2016 – 2025 (\$MILLION)

Country	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
United Kingdom	124	140	158	178	201	227	257	290	328	370	13.0%
France	94.0	105	118	133	149	167	188	211	237	267	12.3%
Germany	146	166	189	215	245	279	318	362	412	470	13.9%
Russia	90.2	101	114	128	144	162	183	206	232	261	12.6%
Rest of Europe	200	228	259	295	335	382	435	495	564	642	13.9%
Total	655	741	838	949	1,075	1,217	1,379	1,563	1,772	2,010	13.3%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.3.3 Europe Market Size & Forecast, By Type

TABLE 29 EUROPE UNDERWATER DRONE MARKET SHARE VALUE, BY TYPE, 2016 – 2025
(\$MILLION)

Type	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
ROV	487	541	602	669	744	828	921	1,025	1,142	1,272	11.3%
AUV	168	200	237	280	331	390	458	538	631	738	17.5%
Total	655	741	838	949	1,075	1,217	1,379	1,563	1,772	2,010	13.3%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.3.4 Europe Market Size & Forecast, By Application

TABLE 30 EUROPE UNDERWATER DRONE MARKET SHARE VALUE, BY APPLICATION, 2016 – 2025
(\$MILLION)

Application	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR%(2019–25)
Commercial Exploration	336	382	434	493	560	636	724	823	937	1,066	13.7%
Defense	132	148	166	186	209	234	263	295	331	372	12.2%
Scientific Research	175	199	225	255	289	328	372	423	480	545	13.5%
Others	10.9	12.1	13.5	15.1	16.8	18.6	20.6	22.8	25.1	27.7	10.6%
Total	655	741	838	949	1,075	1,217	1,379	1,563	1,772	2,010	13.3%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.3.5 Europe Market Size & Forecast, By Component

TABLE 31 EUROPE UNDERWATER DRONE MARKET SHARE VALUE, BY COMPONENT, 2016 – 2025 (\$MILLION)

Component	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
Sensors	122	140	159	182	208	238	272	310	355	406	14.3%
Synthetic Aperture Sonar	39.3	44.4	50.2	56.8	64.2	72.6	82.1	92.9	105	119	13.1%
Inertial Navigation Systems	70.2	79.5	90.2	102	116	132	149	170	193	219	13.5%
Imaging Systems	222	251	284	321	363	411	465	527	598	678	13.3%
Lighting Systems	103	117	132	150	169	192	218	247	280	317	13.4%
Others	97.8	110	123	138	154	173	193	216	242	270	11.9%
Total	655	741	838	949	1,075	1,217	1,379	1,563	1,772	2,010	13.3%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.3.6 Europe Market Size & Forecast, By Propulsion System

TABLE 32 EUROPE UNDERWATER DRONE MARKET SHARE VALUE, BY PROPULSION SYSTEM, 2016 – 2025 (\$MILLION)

Propulsion System	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
Electric System	383	434	492	558	633	719	816	927	1,053	1,197	13.5%
Mechanical System	107	119	134	150	169	190	213	239	269	302	12.3%
Hybrid Systems	165	187	212	240	272	309	350	397	450	511	13.4%
Total	655	741	838	949	1,075	1,217	1,379	1,563	1,772	2,010	13.3%

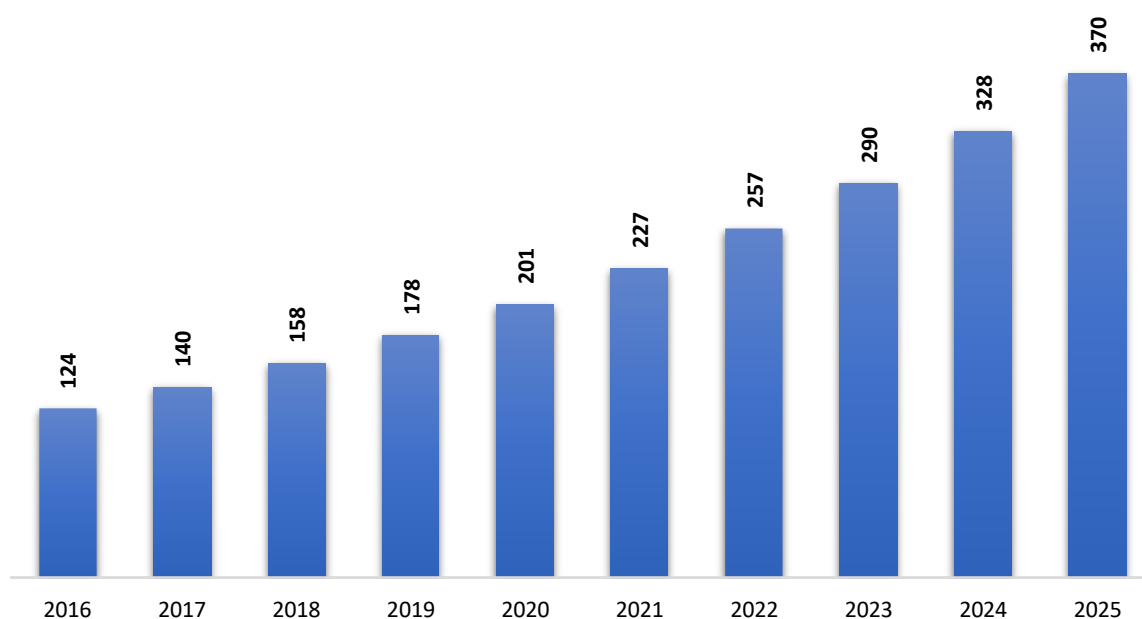
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.3.7 UK

United Kingdom Underwater Drone market was valued at \$158 million in 2018 and is expected to reach a value of \$370 million by the end of 2025, growing at a CAGR of 13.0% from 2019 to 2025.

8.3.7.1 Market Size and Forecast

FIGURE 10 UK UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



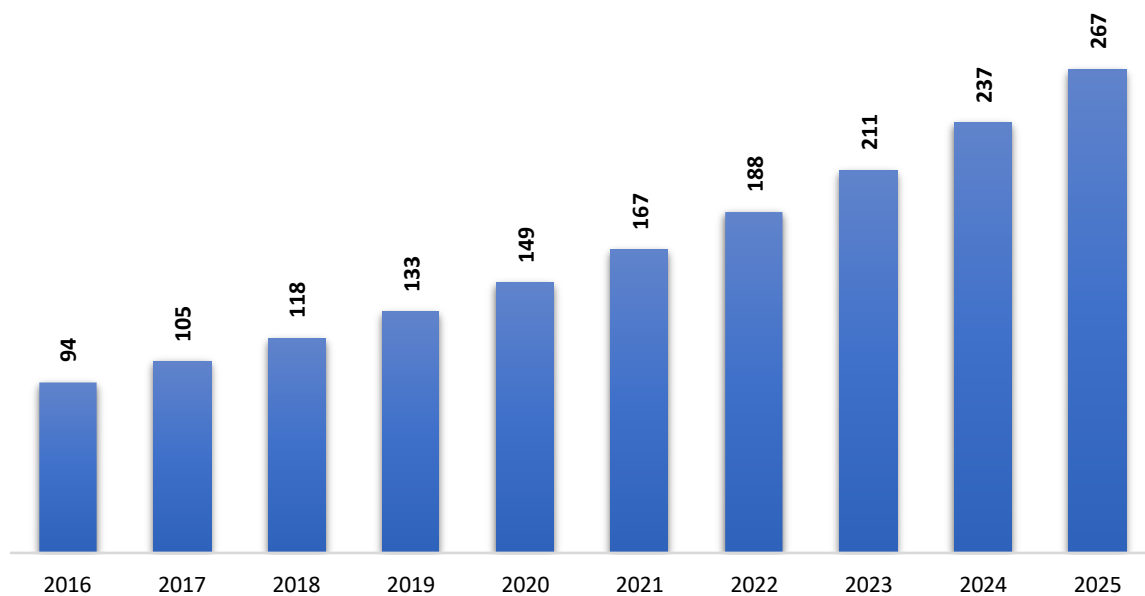
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.3.8 France

France Underwater Drone market was valued at \$118 million in 2018 and is expected to reach a value of \$267 million by the end of 2025, growing at a CAGR of 12.3% from 2019 to 2025.

8.3.8.1 Market Size and Forecast

FIGURE 11 FRANCE UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



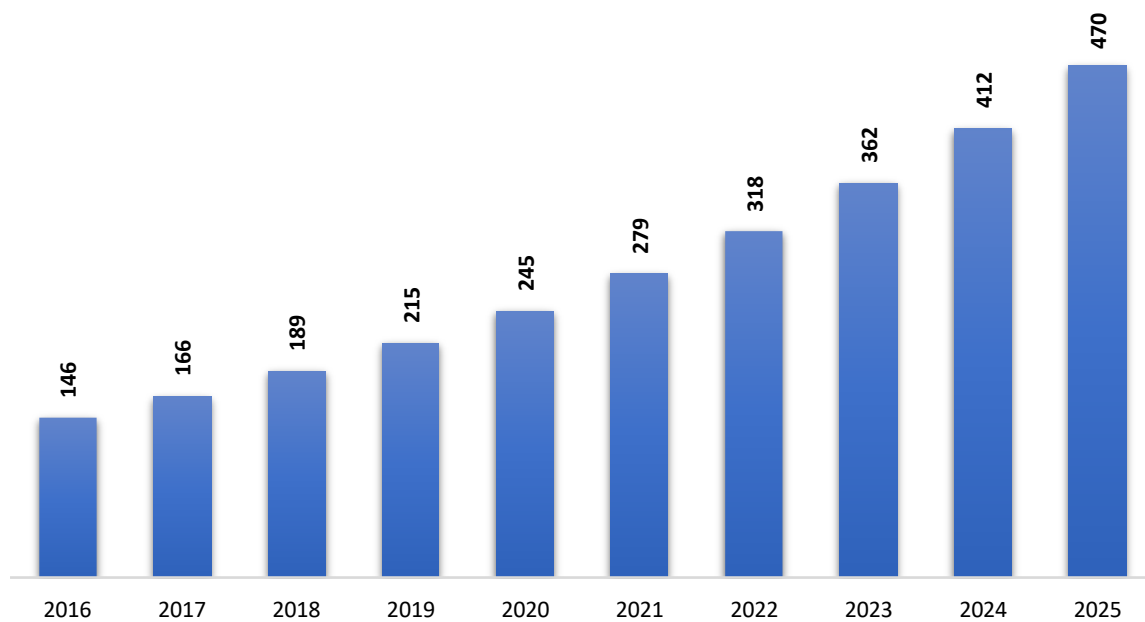
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.3.9 Germany

Germany Underwater Drone market was valued at \$189 million in 2018 and is expected to reach a value of \$470 million by the end of 2025, growing at a CAGR of 13.9% from 2019 to 2025.

8.3.9.1 Market Size and Forecast

FIGURE 12 GERMANY UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



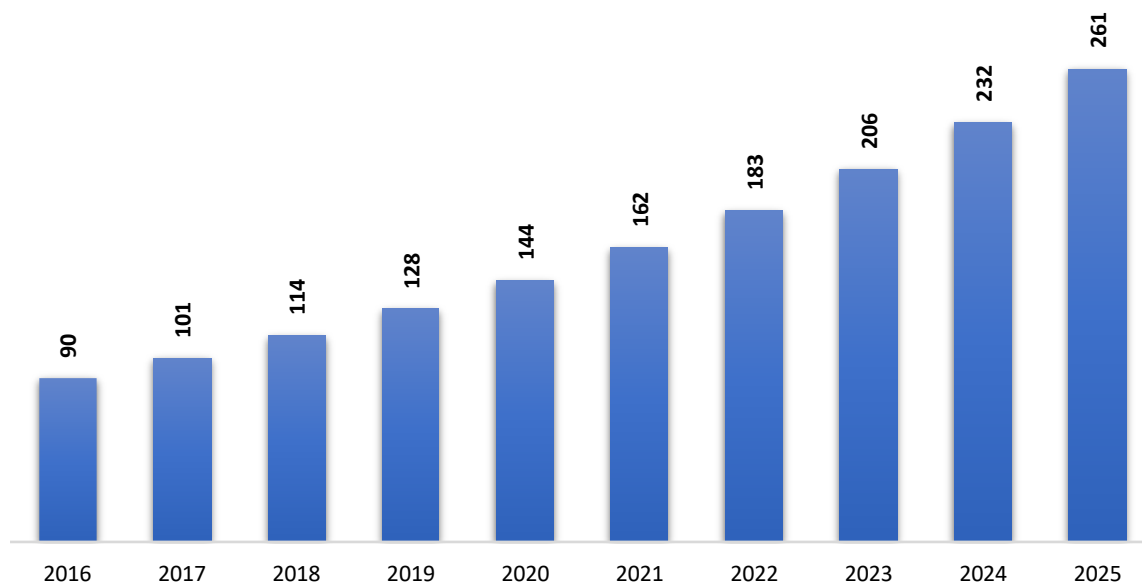
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.3.10 Russia

Russia Underwater Drone market was valued at \$114 million in 2018 and is expected to reach a value of \$261 million by the end of 2025, growing at a CAGR of 12.6% from 2019 to 2025.

8.3.10.1 Market Size and Forecast

FIGURE 13 RUSSIA UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



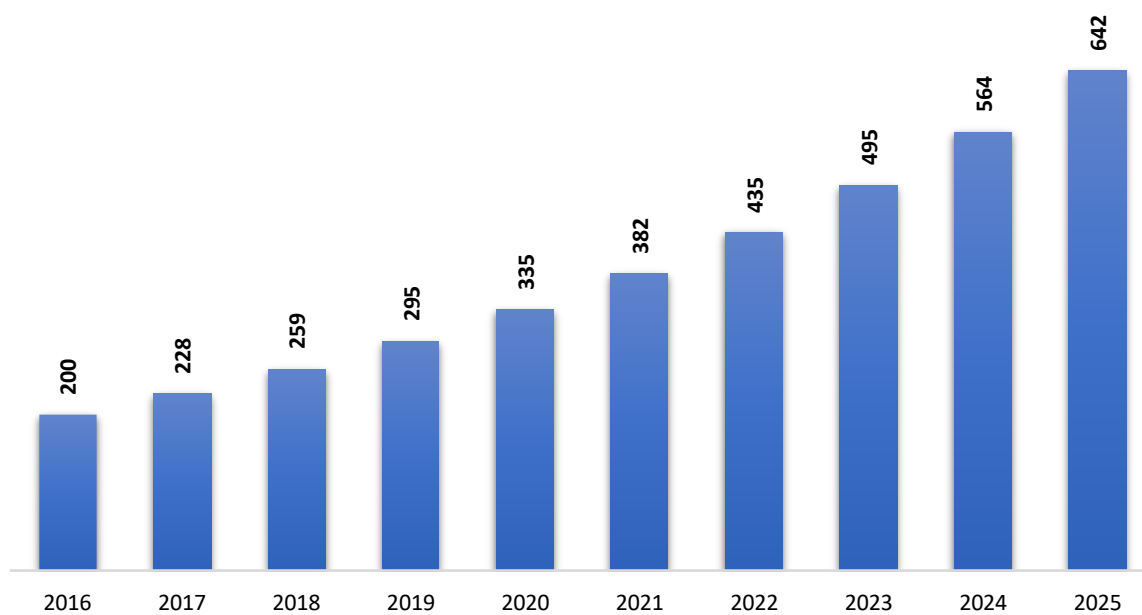
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.3.11 Rest of Europe

Rest of Europe Underwater Drone market was valued at \$259 million in 2018 and is expected to reach a value of \$642 million by the end of 2025, growing at a CAGR of 13.9% from 2019 to 2025.

8.3.11.1 Market Size and Forecast

FIGURE 14 REST OF EUROPE UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



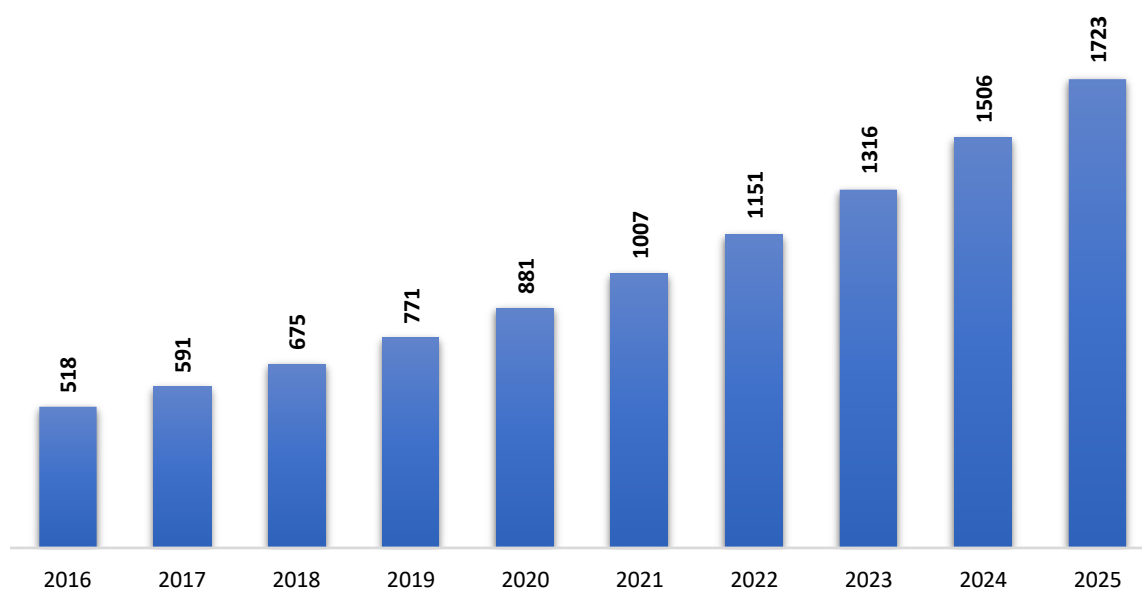
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.4 Asia-Pacific

The rise in federal spending towards naval deployment of underwater vehicles to monitor military threats, commercialization of technologically advanced marine underwater drone solutions, and rise in product adoption across demand-rich countries. The presence of emerging superpowers such as China, India, and Japan, with high-spending power and expansive market opportunities, is expected to further propel the regional market towards growth. For instance, November 05, 2018, Japan announced its intentions to introduce unmanned underwater vehicles into territorial waters to step up monitoring activities. The deployment is part of revisions to the defense ministry's medium-term defense plan. Further, on August 14, 2019, Shenzhen-based, Chinese UAV provider Geneinno Technology has appointed Drone Solutions as the company's new tech distributor for Southeast Asia. The Singaporean company will market, sell and provide after-sales service for Geneinno's underwater drones with an emphasis on the Poseidon model. Moreover, on September 18, 2018, India-based start-up IROV Technologies Pvt Ltd announced that it is launching commercial underwater drones by handing over the first vehicle to Naval Physical and Oceanographic Laboratory (NPOL), a laboratory of the Indian Defence Research and Development Organisation (DRDO).

8.4.1 Market Size & Forecast

TABLE 33 ASIA-PACIFIC, UNDERWATER DRONE MARKET, 2016 – 2025 (\$MILLION)



Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.4.2 Asia-Pacific Market Size & Forecast, By Country

TABLE 34 ASIA-PACIFIC UNDERWATER DRONE MARKET SHARE VALUE, BY COUNTRY, 2016 – 2025 (\$MILLION)

Country	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
China	166	191	218	250	287	329	378	434	498	572	14.7%
India	85.8	98.3	113	129	148	170	194	223	256	294	14.7%
Japan	94.6	108	122	139	159	180	206	234	267	304	13.9%
South Korea	78.3	89.5	102	117	134	153	175	200	229	262	14.4%
Rest of Asia-pacific	92.9	105	120	136	154	175	199	226	257	292	13.6%
Total	518	591	675	771	881	1,007	1,151	1,316	1,506	1,723	14.3%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.4.3 Asia-Pacific Market Size & Forecast, By Type

TABLE 35 ASIA-PACIFIC UNDERWATER DRONE MARKET SHARE VALUE, BY TYPE, 2016 – 2025
(\$MILLION)

Type	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
ROV	395	442	494	552	618	691	774	867	971	1,088	12.0%
AUV	123	150	181	219	263	316	377	450	535	635	19.4%
Total	518	591	675	771	881	1,007	1,151	1,316	1,506	1,723	14.3%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.4.4 Asia-Pacific Market Size & Forecast, By Application

TABLE 36 ASIA-PACIFIC UNDERWATER DRONE MARKET SHARE VALUE, BY APPLICATION, 2016 – 2025 (\$MILLION)

Application	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
Commercial Exploration	278	319	365	418	479	548	629	721	827	948	14.6%
Defense	105	119	135	153	174	197	224	255	289	329	13.6%
Scientific Research	125	143	164	188	215	247	283	325	372	427	14.6%
Others	9.8	10.7	11.6	12.6	13.6	14.6	15.6	16.7	17.6	18.5	6.7%
Total	518	591	675	771	881	1,007	1,151	1,316	1,506	1,723	14.3%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.4.5 Asia-Pacific Market Size & Forecast, By Component

TABLE 37 ASIA-PACIFIC UNDERWATER DRONE MARKET SHARE VALUE, BY COMPONENT, 2016 – 2025 (\$MILLION)

Component	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
Sensors	93.9	108	125	144	166	191	220	254	294	339	15.4%
Synthetic Aperture Sonar	30.2	34.4	39.2	44.7	51.0	58.2	66.4	75.9	86.6	99.0	14.2%
Inertial Navigation Systems	53.9	61.6	70.5	80.6	92.3	106	121	139	159	182	14.5%
Imaging Systems	170	194	222	253	289	330	377	431	493	564	14.3%
Lighting Systems	79.1	90.3	103	118	135	154	176	202	231	264	14.4%
Others	90.6	103	116	132	149	168	190	215	244	275	13.1%
Total	518	591	675	771	881	1,007	1,151	1,316	1,506	1,723	14.3%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.4.6 Asia-Pacific Market Size & Forecast, By Propulsion System

TABLE 38 ASIA-PACIFIC UNDERWATER DRONE MARKET SHARE VALUE, BY PROPULSION SYSTEM, 2016 – 2025 (\$MILLION)

Propulsion System	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
Electric System	298	341	390	447	512	587	673	772	886	1,016	14.7%
Mechanical System	87.5	98.6	111	125	142	160	180	204	230	260	12.9%
Hybrid Systems	133	152	174	199	227	260	298	341	390	447	14.5%
Total	518	591	675	771	881	1,007	1,151	1,316	1,506	1,723	14.3%

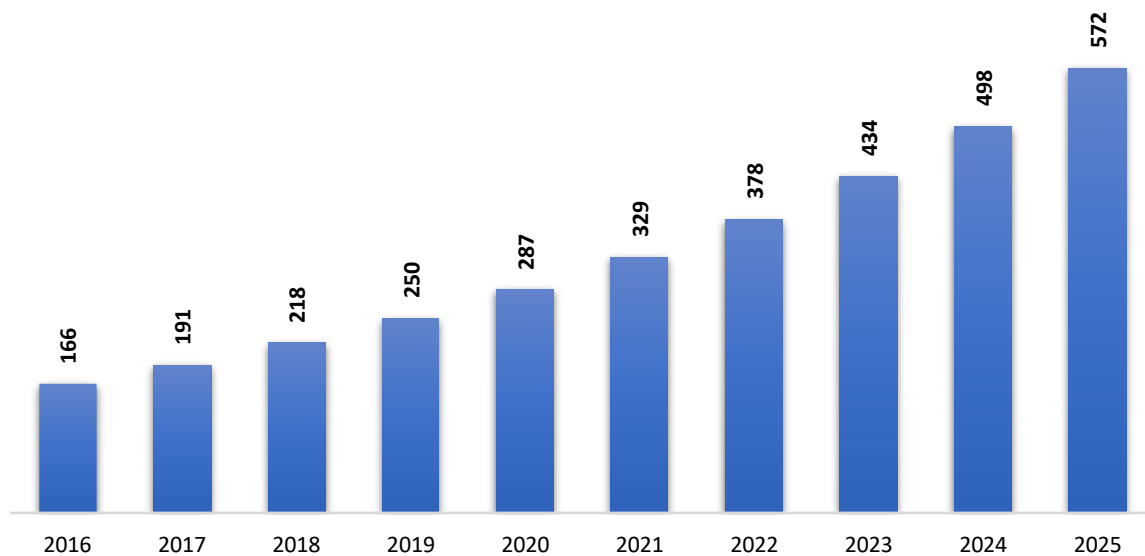
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.4.7 China

China Underwater Drone market was valued at \$218 million in 2018 and is expected to reach a value of \$572 million by the end of 2025, growing at a CAGR of 14.7% from 2019 to 2025.

8.4.7.1 Market Size and Forecast

FIGURE 15 CHINA UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



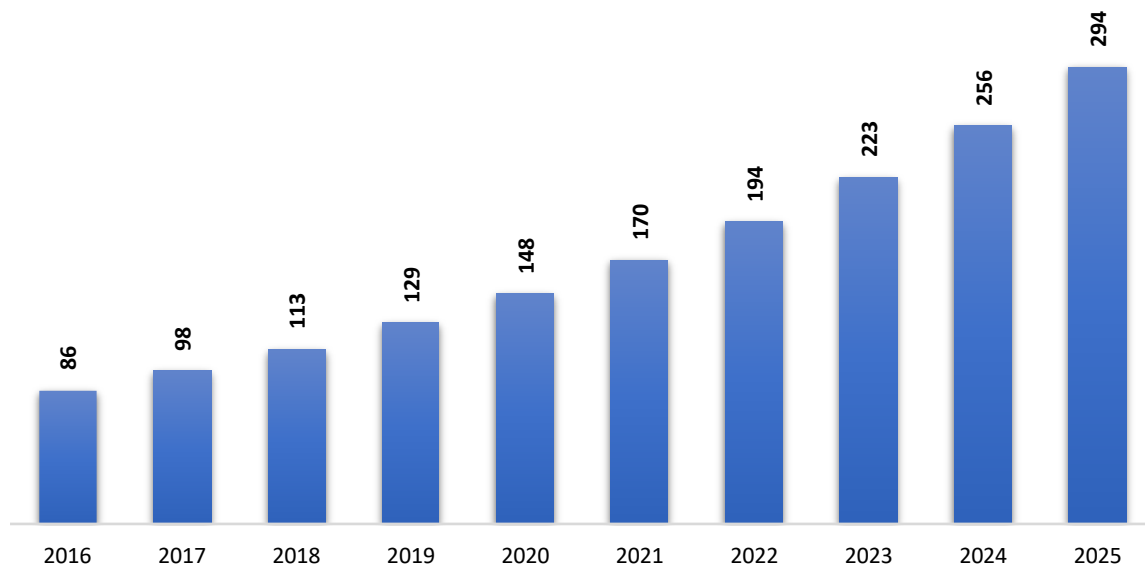
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.4.8 India

India Underwater Drone market was valued at \$113 million in 2018 and is expected to reach a value of \$294 million by the end of 2025, growing at a CAGR of 14.7% from 2019 to 2025.

8.4.8.1 Market Size and Forecast

FIGURE 16 INDIA UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



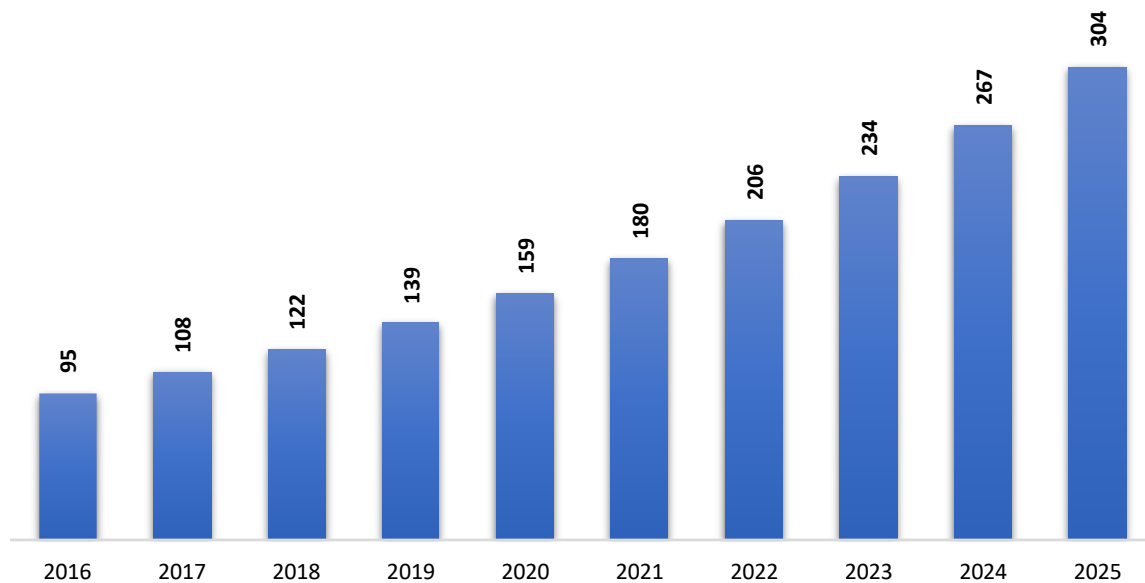
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.4.9 Japan

Japan Underwater Drone market was valued at \$122 million in 2018 and is expected to reach a value of \$304 million by the end of 2025, growing at a CAGR of 13.9% from 2019 to 2025.

8.4.9.1 Market Size and Forecast

FIGURE 17 JAPAN UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



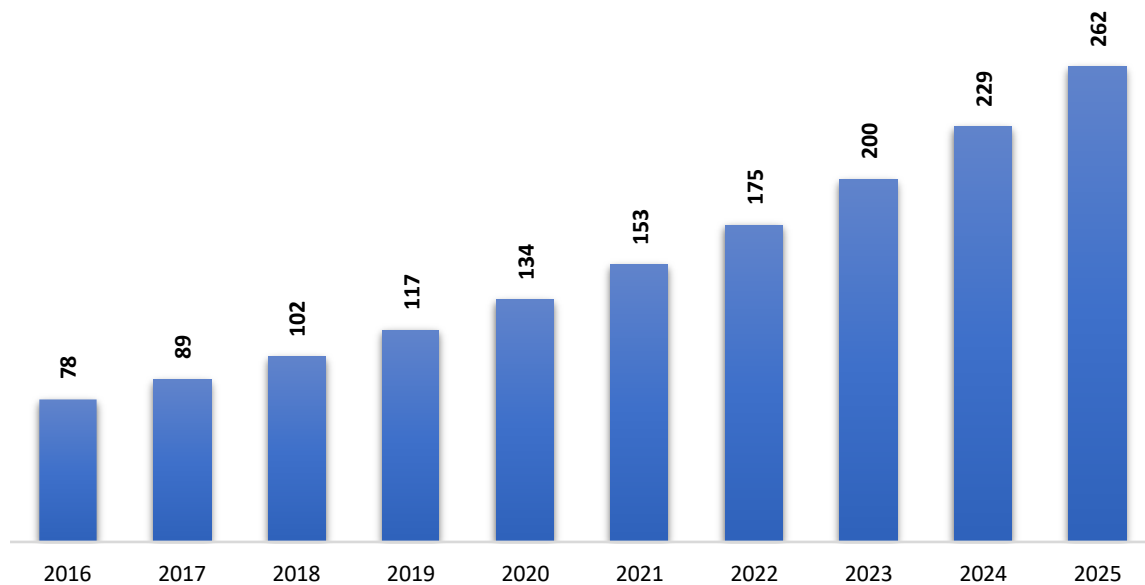
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.4.10 South Korea

South Korea Underwater Drone market was valued at \$102 million in 2018 and is expected to reach a value of \$262 million by the end of 2025, growing at a CAGR of 14.4% from 2019 to 2025.

8.4.10.1 Market Size and Forecast

FIGURE 18 SOUTH KOREA UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



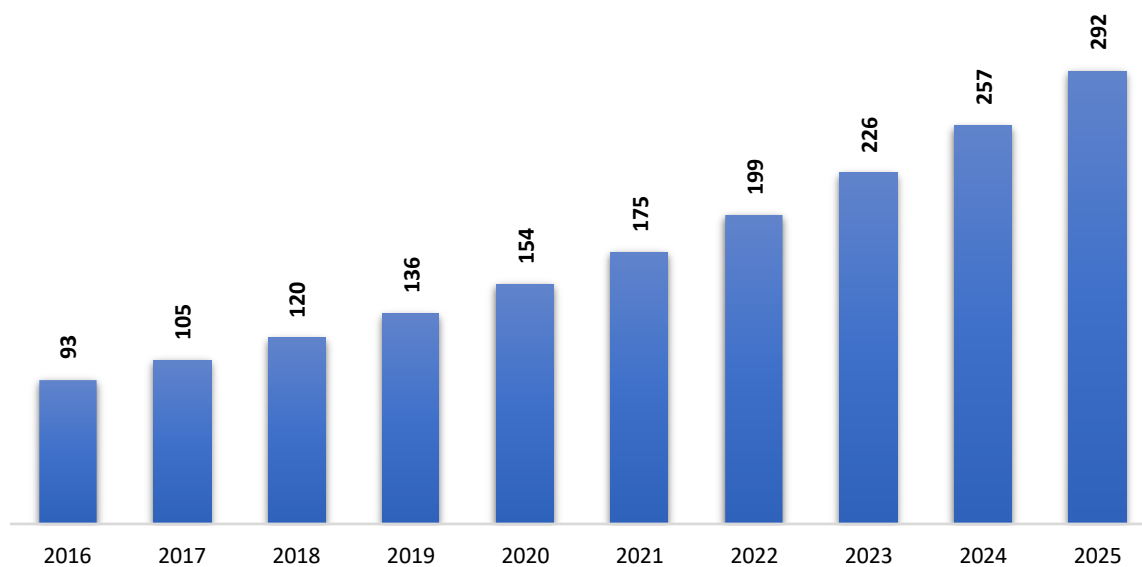
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.4.11 Rest of Asia-Pacific

Rest of Asia-Pacific Underwater Drone market was valued at \$120 million in 2018 and is expected to reach a value of \$292 million by the end of 2025, growing at a CAGR of 13.6% from 2019 to 2025.

8.4.11.1 Market Size and Forecast

FIGURE 19 REST OF ASIA-PACIFIC UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



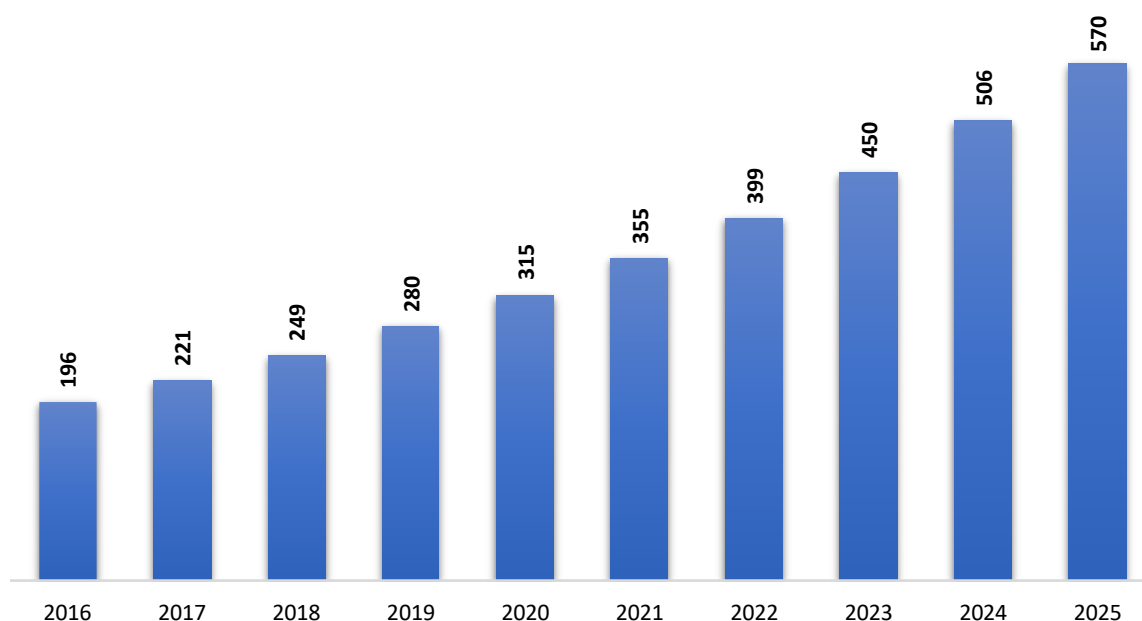
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.5 Rest of the World

The Rest of the World (RoW), consisting of countries from the Middle East, Africa, and South America region, presents untapped opportunities and market potential for several key manufacturers. The gradual development of underwater drones for several key applications in the oil & gas sector, offshore energy, military, and underwater life sciences research, would substantially improve the overall growth prospects of the Underwater Drone in the region. The companies operating in the region are thus focused on expansion of business activities and strengthening of regional business position. For instance, on July 26, 2017, Teledyne Gavia announced that it had supplied a Gavia Offshore Surveyor AUV system to the Saudi Arabian Oil Company (Saudi Aramco). The Gavia AUV system is expected to enhance Saudi Aramco's agility in meeting high demands for hydrographic survey services through the company's in-house survey operations. The Gavia system will be utilized predominantly for debris survey and pipeline inspection in the Arabian Gulf, with a potential service extension for other type of surveys in water depths up to 1000m in the Red Sea region.

8.5.1 Market Size & Forecast

TABLE 39 REST OF THE WORLD, UNDERWATER DRONE MARKET, 2016 – 2025 (\$MILLION)



Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.5.2 Rest of the World Market Size & Forecast, By Country

TABLE 40 REST OF THE WORLD UNDERWATER DRONE MARKET SHARE VALUE, BY COUNTRY, 2016 – 2025 (\$MILLION)

Country	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
South America	111	126	142	161	182	207	234	265	301	341	13.3%
Middle East	66.7	74.4	83.2	92.8	104	116	129	144	161	180	11.7%
Africa	18.6	20.8	23.2	25.9	28.9	32.2	35.8	39.9	44.4	49.4	11.3%
Total	196	221	249	280	315	355	399	450	506	570	12.6%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.5.3 Rest of the World Market Size & Forecast, By Type

TABLE 41 REST OF THE WORLD UNDERWATER DRONE MARKET SHARE VALUE, BY TYPE, 2016 – 2025 (\$MILLION)

Type	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
ROV	146	162	180	199	221	246	273	303	336	373	11.0%
AUV	50.2	58.9	68.9	80.5	93.8	109	127	147	171	197	16.1%
Total	196	221	249	280	315	355	399	450	506	570	12.6%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.5.4 Rest of the World Market Size & Forecast, By Application

TABLE 42 REST OF THE WORLD UNDERWATER DRONE MARKET SHARE VALUE, BY APPLICATION, 2016 – 2025 (\$MILLION)

Application	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
Commercial Exploration	102	115	130	147	166	188	212	239	270	305	12.9%
Defense	39.6	44.1	49.3	54.9	61.3	68.3	76.2	85.1	94.9	106	11.6%
Scientific Research	51.8	58.4	65.9	74.3	83.8	94.6	107	120	136	153	12.8%
Others	2.6	2.9	3.2	3.5	3.8	4.2	4.5	4.9	5.3	5.7	8.4%
Total	196	221	249	280	315	355	399	450	506	570	12.6%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.5.5 Rest of the World Market Size & Forecast, By Component

TABLE 43 REST OF THE WORLD UNDERWATER DRONE MARKET SHARE VALUE, BY COMPONENT, 2016 – 2025 (\$MILLION)

Component	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
Sensors	36.3	41.2	46.8	53.2	60.4	68.6	77.9	88.6	101	114	13.6%
Synthetic Aperture Sonar	11.7	13.1	14.7	16.6	18.6	20.9	23.5	26.4	29.7	33.4	12.4%
Inertial Navigation Systems	20.8	23.5	26.5	29.9	33.7	38.0	42.8	48.3	54.5	61.5	12.8%
Imaging Systems	65.8	74.0	83.2	93.6	105	118	133	150	169	190	12.6%
Lighting Systems	30.5	34.4	38.8	43.6	49.1	55.4	62.4	70.2	79.1	89.1	12.6%
Others	31.0	34.6	38.6	43.0	47.9	53.3	59.3	66.0	73.4	81.5	11.2%
Total	196	221	249	280	315	355	399	450	506	570	12.6%

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.5.6 Rest of the World Market Size & Forecast, By Propulsion System

TABLE 44 REST OF THE WORLD UNDERWATER DRONE MARKET SHARE VALUE, BY PROPULSION SYSTEM, 2016 – 2025 (\$MILLION)

Propulsion System	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	CAGR% (2019–25)
Electric System	116	131	148	167	189	213	240	271	306	345	12.8%
Mechanical System	31.1	34.7	38.8	43.3	48.3	53.9	60.1	67.1	74.9	83.6	11.6%
Hybrid Systems	48.5	54.7	61.6	69.4	78.2	88.1	99.3	112	126	142	12.7%
Total	196	221	249	280	315	355	399	450	506	570	12.6%

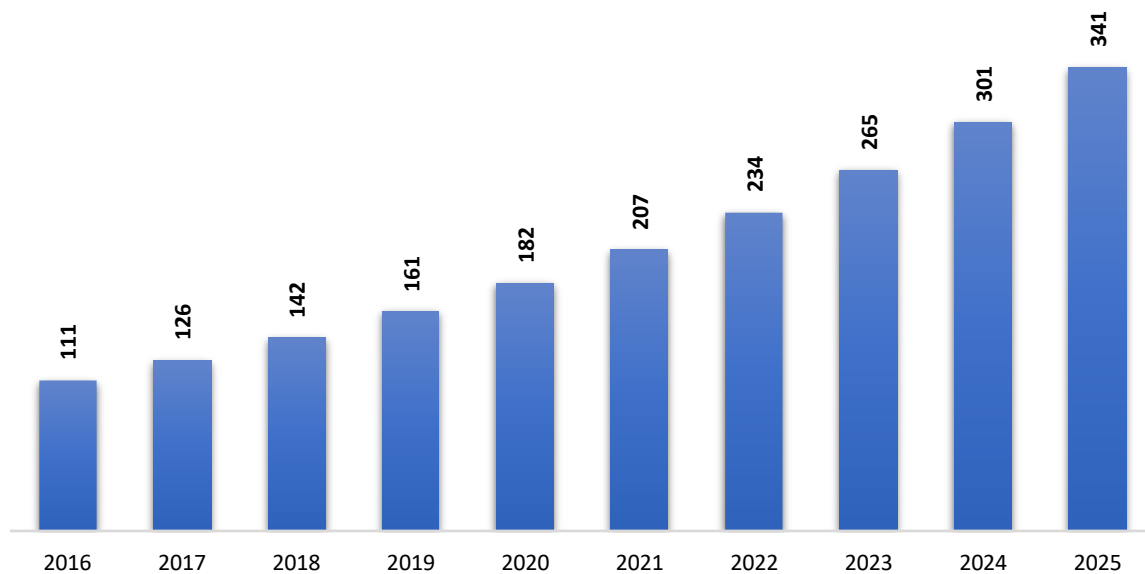
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.5.7 South America

South America Underwater Drone market was valued at \$142 million in 2018 and is expected to reach a value of \$341 million by the end of 2025, growing at a CAGR of 13.3% from 2019 to 2025.

8.5.7.1 Market Size and Forecast

FIGURE 20 SOUTH AMERICA UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



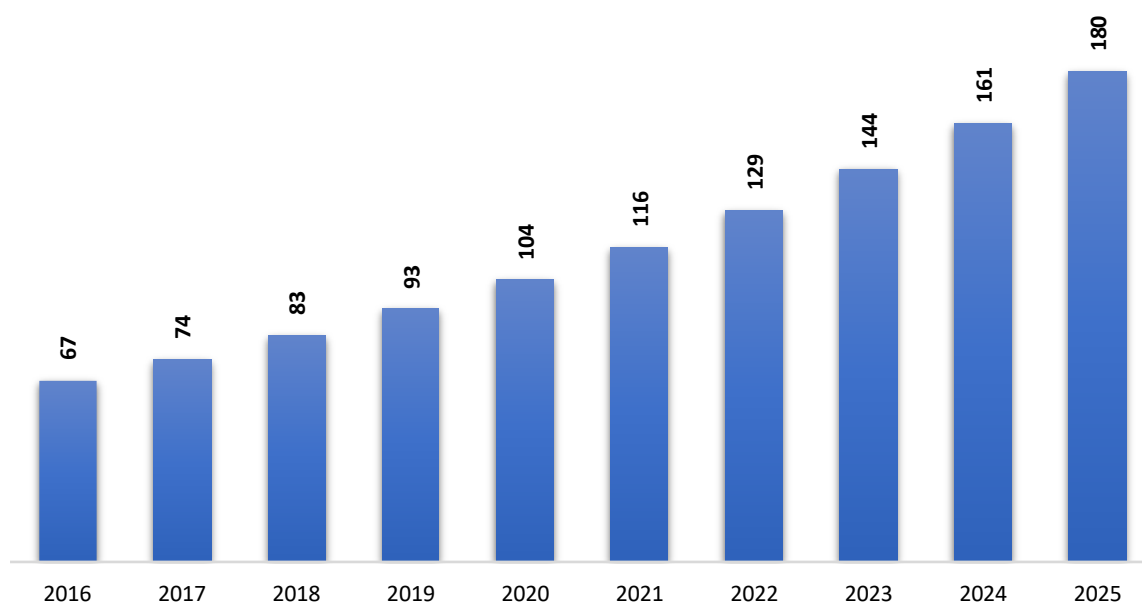
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.5.8 Middle East

Middle East Underwater Drone market was valued at \$83.2 million in 2018 and is expected to reach a value of \$180 million by the end of 2025, growing at a CAGR of 11.7% from 2019 to 2025.

8.5.8.1 Market Size and Forecast

FIGURE 21 MIDDLE EAST UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



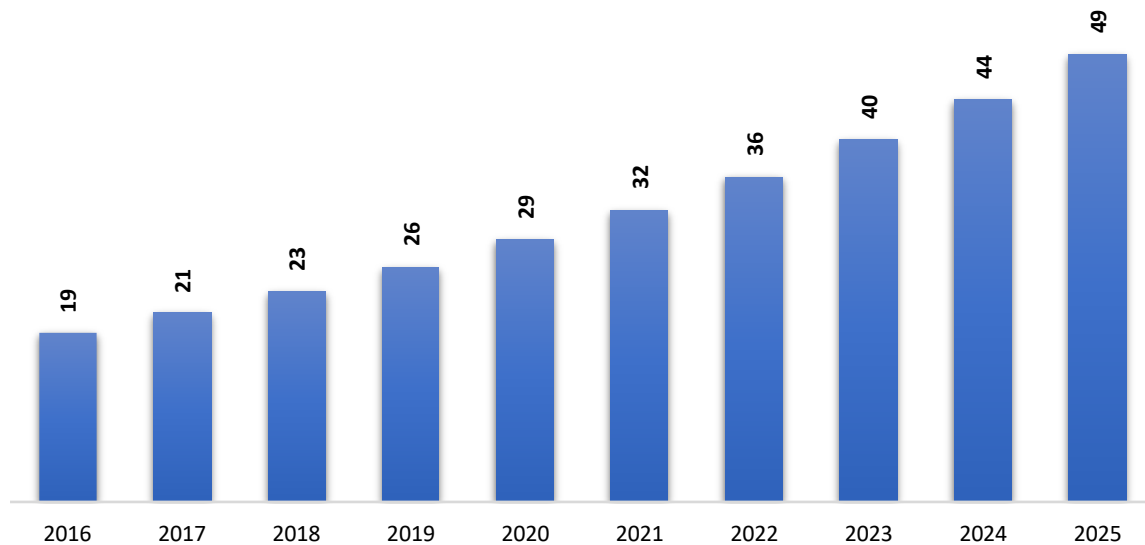
Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

8.5.9 Africa

Africa Underwater Drone market was valued at \$23.2 million in 2018 and is expected to reach a value of \$49.4 million by the end of 2025, growing at a CAGR of 11.3% from 2019 to 2025.

8.5.9.1 Market Size and Forecast

FIGURE 22 AFRICA UNDERWATER DRONE MARKET SHARE, 2016-2025 (\$MILLION)



Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

9 Company Profile

9.1 Kongsberg Gruppen

Kirkegårdsveien 45

Kongsberg 3616

Norway

9.1.1 Company Basic Information

TABLE 45 KONGSBERG GRUPPEN BASIC INFORMATION AND ITS COMPETITORS

#	Item	Description
1	Company Name	Kongsberg Gruppen
2	Website	www.kongsberg.com
3	Primary Business	Aerospace & Defense
4	Business Segments	Kongsberg Defence & Aerospace, Kongsberg Maritime, and Other
5	Key Executives	Geir Haoy- President and Chief Executive Officer
6	Key Strategies	Business Development
7	Competitors	Oceaneering International, Inc., Lockheed Martin Corporation, Teledyne Technologies, Inc., Saab Group, Blueye, Aquabotix Technology Corporation, The Boeing Company, Fugro N.V., Subsea 7 S.A.

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

9.1.2 Kongsberg Gruppen Product Category, Application, and Specification

9.1.2.1 HUGIN

HUGIN autonomous underwater vehicles can function without tethers, cables, or remote control, they have a multitude of applications in oceanography, environmental monitoring, and underwater resource studies. HUGIN AUVs are equipped with acoustic communication links providing real time status monitoring of the vehicle and payload sensors.

9.1.2.2 MUNIN

MUNIN has a comprehensive navigation and positioning capability that can tie the data to a meaningful georeferenced solution. The capability is further enhanced by the autonomous pipe tracking software, making survey and inspection work easy for operators.

9.1.2.3 EELUME

Eelume underwater Intervention vehicles are self-propelled robotic arms whose slender and flexible body can transit over long distances and carry out light inspection, maintenance and repair jobs in confined spaces. It is equipped with several types of sensors and tools needed to fulfil its mission.

9.1.2.4 SEAGLIDER

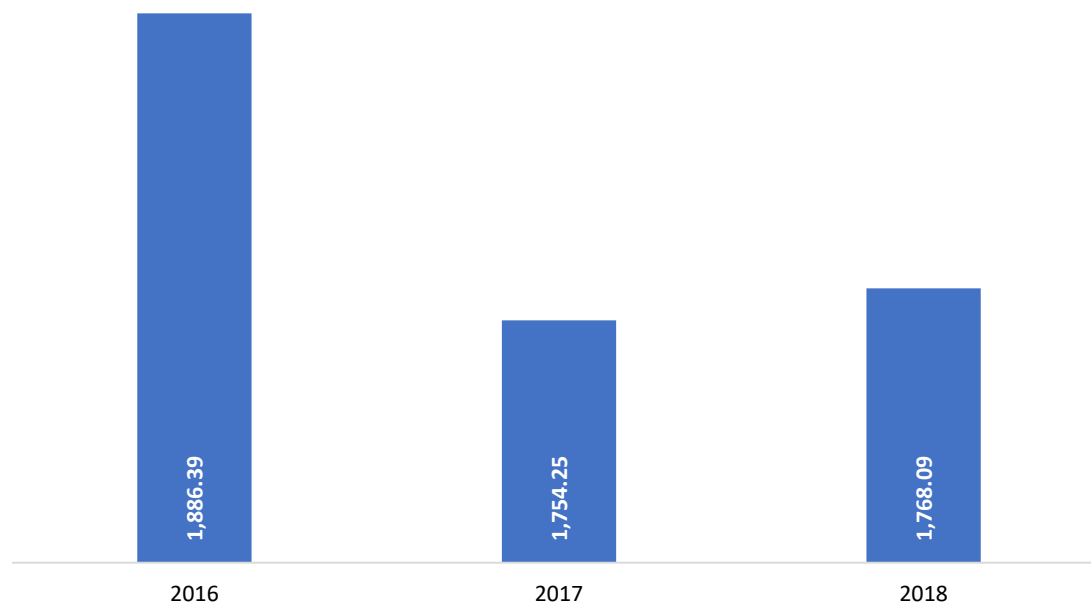
Seagliders are a range of autonomous underwater vehicles (AUV) developed for continuous, long term measurement of oceanographic parameters.

9.1.2.5 REMUS100

The REMUS 100 AUV include a wide variety of standard and customer specified sensors and system options. REMUS 100 is suitable for marine research, defence, hydrographic and offshore/energy markets. It contains sophisticated sensor, navigation, and power resources to perform intricate sonar and oceanographic surveys over large areas. The REMUS range of products also include REMUS M3V, REMUS 600, and REMUS 6000.

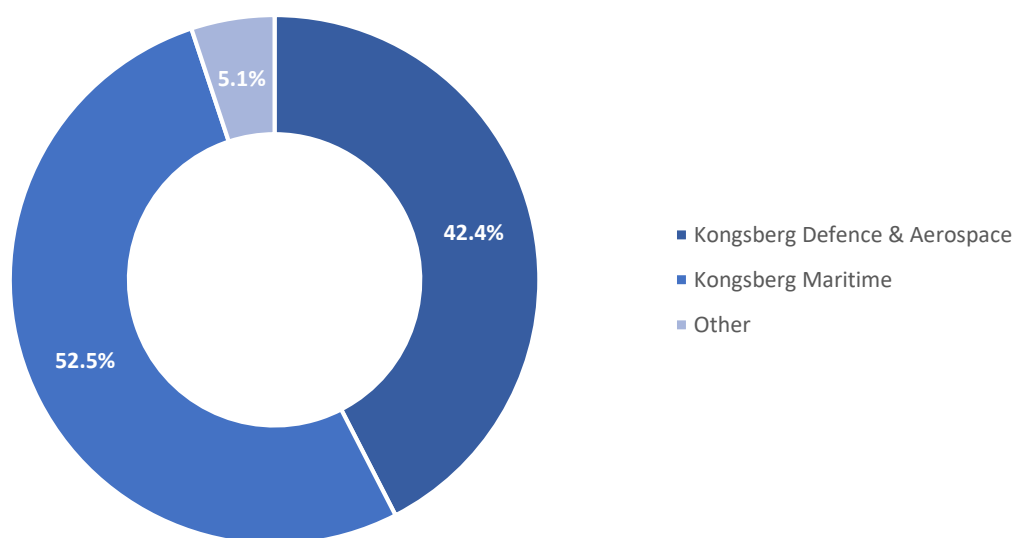
9.1.3 Kongsberg Gruppen Financial Performance (2016-2018)

FIGURE 23 KONGSBERG GRUPPEN REVENUE, 2016-2018, (\$MILLION)



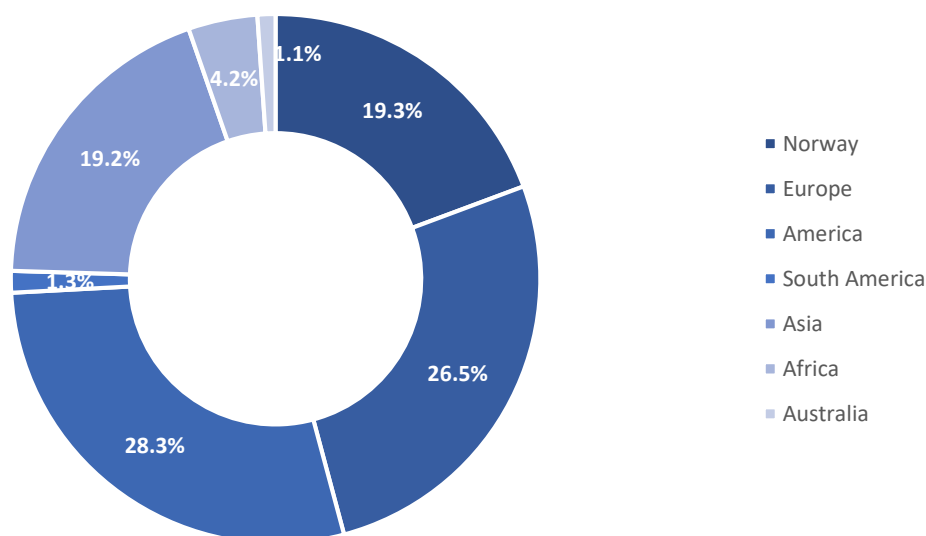
Source: Secondary Sources, Company Publications, Primary Interviews, and ERC Analysis

FIGURE 24 KONGSBERG GRUPPEN REVENUE, BY BUSINESS SEGEMENT, 2018 (%)



Source: Secondary Sources, Company Publications, Primary Interviews, and ERC Analysis

FIGURE 25 KONGSBERG GRUPPEN REVENUE, BY REGION, 2018 (%)



Source: Secondary Sources, Company Publications, Primary Interviews, and ERC Analysis

9.1.4 Kongsberg Gruppen Recent Development

TABLE 46 RECENT DEVELOPMENTS

Date	Type	Description
November 26, 2018	Business Development	Hydroid, Inc., a subsidiary of Kongsberg Maritime and a leading manufacturer of autonomous underwater vehicles (AUVs), have received an order for a New Generation REMUS 6000 AUV from the Japan Agency for Marine-Earth Science Technology (JAMSTEC) as part of the Strategic Innovation Promotion Program (SIP)
November 7, 2018	Business Development	Kongsberg Maritime, have received 3 separate orders for a total of five Kongsberg Maritime HUGIN autonomous underwater vehicle systems from Ocean Infinity, a Hydrographic and Geophysical Survey company.
March 27, 2017	Business Development	Kongsberg Maritime have signed a contract with the Norwegian Defence Materiel Agency (NDMA) for delivery of four complete HUGIN AUV systems (Autonomous Underwater Vehicles) for detection, classification and identification of mines.

Source: Company publications & ERC Analysis

9.1.5 Company Overview

Kongsberg Gruppen ASA was founded in 1814 and is headquartered in Kongsberg, Norway. Kongsberg Gruppen ASA, together with its subsidiaries, provides technology systems and solutions to customers in deep-sea, digital, defence, merchant marine, oil and gas, fisheries, aerospace, and space industries. The company operates in two segments, Kongsberg Maritime, and Kongsberg Defence & Aerospace. The Kongsberg Maritime segment provides products and systems for positioning, monitoring, navigation, and automation for merchant vessels and the offshore industry, as well as back-deck handling equipment for use on offshore vessels. The Kongsberg Defence & Aerospace segment supplies defense and space-related systems and products for command and control, weapons control systems and surveillance, communications solutions, remote weapon systems, and missiles. The company primarily operates in Norway, Europe, North America, South America, Asia, Australia, and Africa.

9.2 Oceaneering International, Inc.

11911 FM 529

Houston, TX 77041

United States

9.2.1 Company Basic Information

TABLE 47 OCEANEERING INTERNATIONAL, INC. BASIC INFORMATION AND ITS COMPETITORS

#	Item	Description
1	Company Name	Oceaneering International, Inc.
2	Website	www.oceaneering.com
3	Primary Business	Oil & Gas Equipment & Services
4	Business Segments	Remotely Operated Vehicles, Subsea Products, Subsea Projects, Asset Integrity, Advanced Technologies
5	Key Executives	Roderick A. Larson - President and Chief Executive Officer
6	Key Strategies	Business Development
7	Competitors	Kongsberg Maritime, Lockheed Martin Corporation, Teledyne Technologies, Inc., Saab Group, Blueye, Aquabotix Technology Corporation, The Boeing Company, Fugro N.V., Subsea 7 S.A.

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

9.2.2 Oceaneering International, Inc. Product Category, Application, and Specification

9.2.2.1 NEXXUS ROV

The NEXXUS ROV raises the bar and exceeds API S53 specifications to solve demanding tooling intervention requirements. It is a high-power, heavy work class ROV system meets the API Standard 53 requirements for BOP intervention. With heavily increased tooling capabilities.

9.2.2.2 Maxximum ROV

The Maxximum ROV is a side entry cage deployed, dual manipulator 270 HP heavy work class ROV. The cage or tether management system (TMS) supplies an additional 110 HP, powers skids, and has thruster control and auto heading. The system employs a microprocessor-based telemetry system to minimize maintenance, decrease setup time, simplify troubleshooting, and provide automated control functions.

9.2.2.3 Millennium Plus

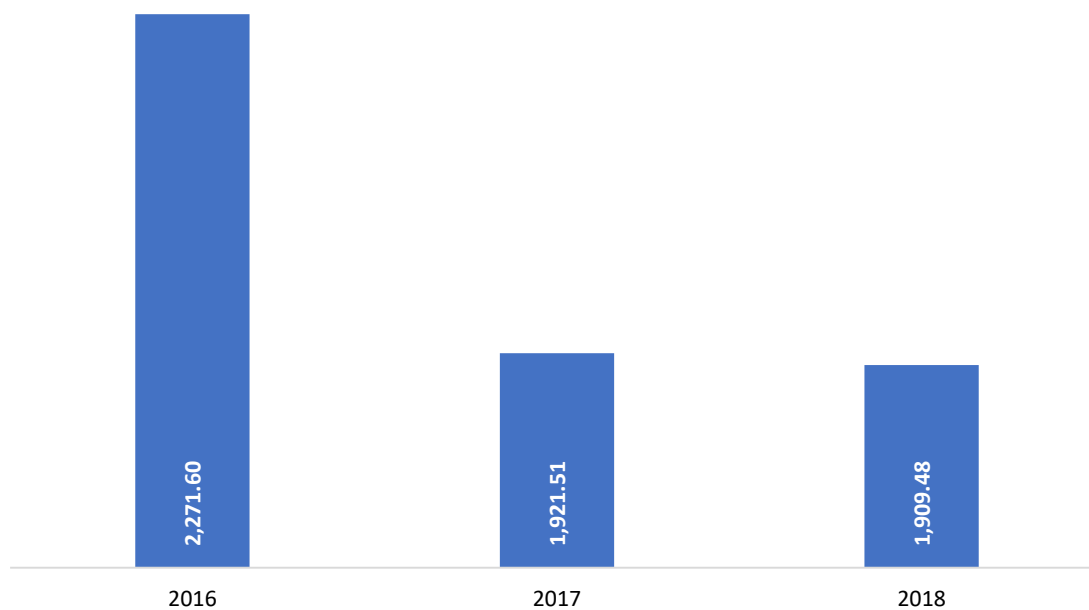
The Millennium Plus ROV is a side entry cage deployed, dual manipulator 220 HP heavy work class ROV. The cage or Tether Management System (TMS) supplies an additional 110 HP and is capable of powering skids and has thruster control and auto heading.

9.2.2.4 Magnum Plus

The Magnum ROV is a side entry cage deployed, dual manipulator ROV that operates in water depths up to 10,000 fsw and in severe weather conditions. The cage or tether management system (TMS) supplies an additional 85hp, powers skids, and has thruster control and auto heading.

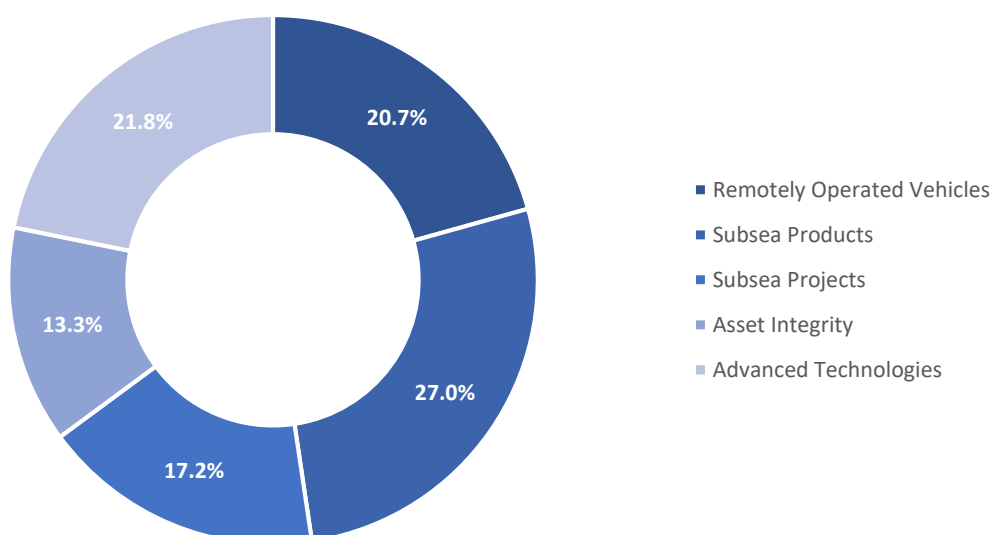
9.2.3 Oceaneering International, Inc. Financial Performance (2016-2018)

FIGURE 26 OCEANEERING INTERNATIONAL, INC. REVENUE, 2016-2018, (\$MILLION)



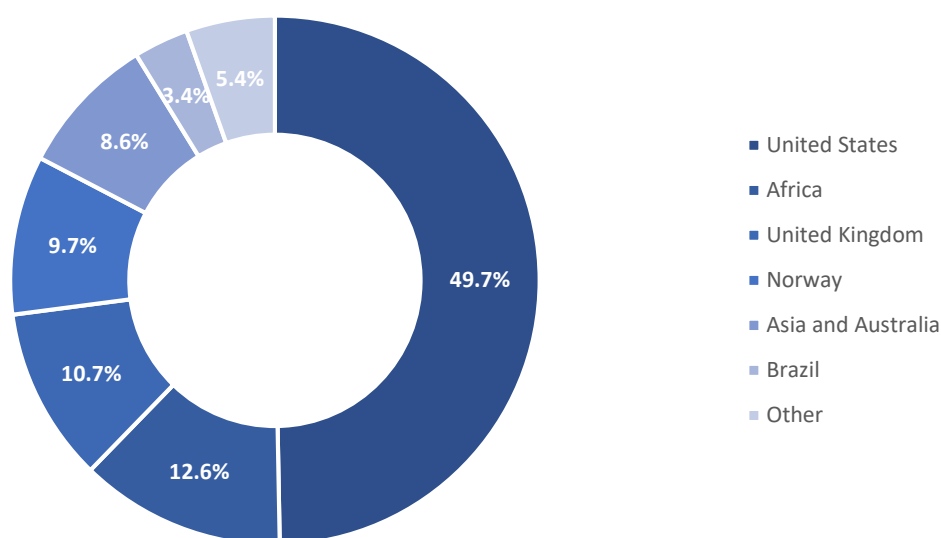
Source: Secondary Sources, Company Publications, Primary Interviews, and ERC Analysis

FIGURE 27 OCEANEERING INTERNATIONAL, INC. REVENUE, BY BUSINESS SEGEMENT, 2018 (%)



Source: Secondary Sources, Company Publications, Primary Interviews, and ERC Analysis

FIGURE 28 OCEANEERING INTERNATIONAL, INC. REVENUE, BY REGION, 2018 (%)



Source: Secondary Sources, Company Publications, Primary Interviews, and ERC Analysis

9.2.4 Oceaneering International, Inc. Recent Development

TABLE 48 RECENT DEVELOPMENTS

Date	Type	Description
Aug. 13, 2018	Business Development	Oceaneering International, Inc. have entered into a contract with Equinor ASA ("Equinor") for a three-year term with two one-year optional extension periods to provide a resident, battery-powered electric remotely operated vehicle ("E-ROV") system to support subsea inspection, maintenance and repair activities. The E-ROV system is expected to be deployed in water depths up to 1000 meters on the Norwegian Continental Shelf.

Source: Company publications & ERC Analysis

9.2.5 Company Overview

Oceaneering International, Inc. was founded in 1964 and is headquartered in Houston, Texas. Oceaneering International, Inc. provides engineered services and products to the offshore oil and gas industry, as well as to defense, aerospace, and commercial theme park industries worldwide. The company's Remotely Operated Vehicles (ROVs) segment offers submersible vehicles for drill support, vessel-based inspection, maintenance and repair, installation and construction support, pipeline inspection and surveys, and subsea production facility operation and maintenance services. As of December 31, 2018, this segment owned 275 work-class ROVs. The company's Subsea Products segment constructs various specialty subsea hardware products, including subsea umbilicals utilizing steel tubes, thermoplastic hoses, and termination assemblies, tooling, ROV tooling, and subsea work packages, production control equipment, installation and workover control systems, clamp connectors, pipeline connector and repair systems, subsea and topside control valves; and subsea chemical injection valves, as well as offers riser-less light well intervention services. Its Subsea Projects segment performs subsea oilfield hardware installation and inspection, maintenance, and repair services, serves shallow water projects; and performs subsea intervention and hardware installation services, such as subsea well tie-backs, pipeline/flow line tie-ins and repairs, pipeline crossing, and umbilical and other subsea equipment installations, and subsea intervention services. The company's Asset Integrity segment offers asset integrity services for the safety of customers' facilities onshore and offshore; third-party inspections to customers in the oil and gas, petrochemical, and power generation industries; and first-pass integrity evaluation and assessment, and nondestructive testing services. Its Advanced Technologies segment provides project management, engineering services, and equipment for applications in non-energy industries.

9.3 Lockheed Martin Corporation

6801 Rockledge Drive

Bethesda, MD 20817

United States

9.3.1 Company Basic Information

TABLE 49 LOCKHEED MARTIN CORPORATION BASIC INFORMATION AND ITS COMPETITORS

#	Item	Description
1	Company Name	Lockheed Martin Corporation
2	Website	www.lockheedmartin.com
3	Primary Business	Aerospace and Defense
4	Business Segments	Aeronautics, Missiles and Fire Control (MFC), Rotary and Mission Systems (RMS), and Space
5	Key Executives	Ms. Marillyn A. Hewson - Chairman, President & CEO
6	Key Strategies	Business Development
7	Competitors	Kongsberg Maritime, Oceaneering International, Inc., Teledyne Technologies, Inc., Saab Group, Blueeye, Aquabotix Technology Corporation, The Boeing Company, Fugro N.V., Subsea 7 S.A.

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

9.3.2 Lockheed Martin Corporation Product Category, Application, and Specification

9.3.2.1 EMATT

EMATT provides combined training of submarine, aircraft, & ship crews in search, detection & tracking of submarines. It's "A-size" Sonobuoy shape means the target can be launched from surface ships or fixed and rotary winged aircraft. EMATT offers advanced ASW training capability with minimal maintenance, training, and logistics to operate and its expendability allows exercises to occur in littoral or blue waters, day or night, and in high or low sea-states.

9.3.2.2 SUBMATT

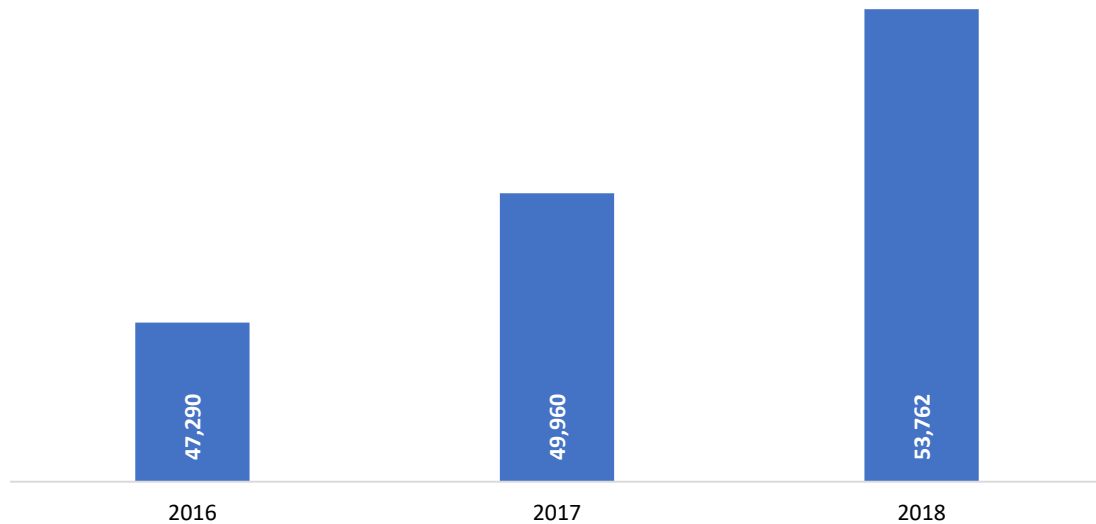
The SUBMATT AAT can store twelve (12) Run Position Switches (RPS) in the target which can be programmed at the LM-Marion factory or created by the user via the PTP.

9.3.2.3 Orca XLUUV

Orca XLUUV is designed to support multiple critical missions. This long-range autonomous vehicle performs a variety of missions. Lockheed Martin's XLUUV leverages its expertise in the integration of proven, and open, architecture solutions.

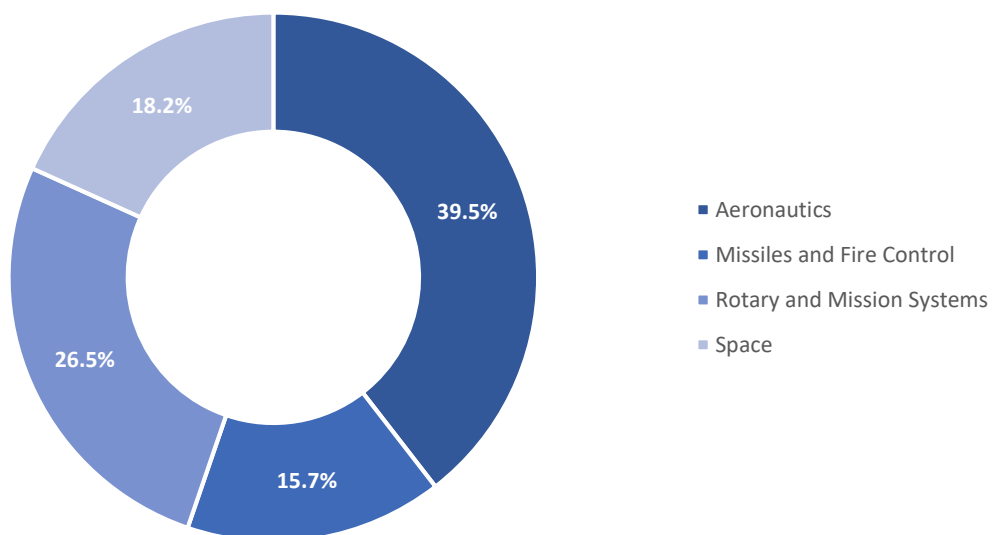
9.3.3 Lockheed Martin Corporation Financial Performance (2016-2018)

FIGURE 29 LOCKHEED MARTIN CORPORATION REVENUE, 2016-2018, (\$MILLION)



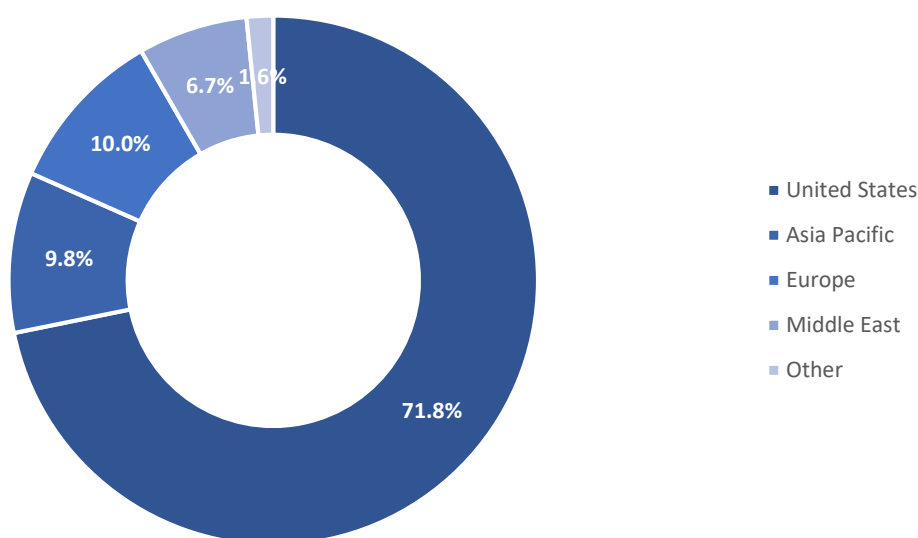
Source: Secondary Sources, Company Publications, Primary Interviews, and ERC Analysis

FIGURE 30 LOCKHEED MARTIN CORPORATION REVENUE, BY BUSINESS SEGEMENT, 2018 (%)



Source: Secondary Sources, Company Publications, Primary Interviews, and ERC Analysis

FIGURE 31 LOCKHEED MARTIN CORPORATION REVENUE, BY REGION, 2018 (%)



Source: Secondary Sources, Company Publications, Primary Interviews, and ERC Analysis

9.3.4 Lockheed Martin Corporation Recent Development

TABLE 50 RECENT DEVELOPMENTS

Date	Type	Description
October 30, 2017	Business Development	Lockheed Martin has received a design-phase contract from the U.S. Navy's for its largest unmanned undersea vehicle valued at \$43.2 million for Orca, the U.S. Navy's Extra Large Unmanned Undersea Vehicle (XLUUV). XLUUV Orca is a two-phase competition, including the currently awarded design phase and a competitive production phase for up to nine vehicles to meet increasing demands for undersea operational awareness and payload delivery.

Source: Company publications & ERC Analysis

9.3.5 Company Overview

Lockheed Martin Corporation was founded in 1995 and is headquartered in Bethesda, Maryland. Lockheed Martin Corporation, a security and aerospace company, engages in the research, design, development, manufacture, integration, and sustainment of technology systems, products, and services worldwide. It operates through four segments: Aeronautics, Missiles and Fire Control (MFC), Rotary and Mission Systems (RMS), and Space. The Aeronautics segment offers combat and air mobility aircraft, unmanned air vehicles, and related technologies. The MFC segment provides air and missile defense systems, tactical missiles and air-to-ground precision strike weapon systems, logistics, fire control systems, mission operations support, readiness, engineering support, and integration services, manned and unmanned ground vehicles, and energy management solutions. The RMS segment offers military and commercial helicopters, ship and submarine mission and combat systems, mission systems and sensors for rotary and fixed-wing aircraft, sea and land-based missile defense systems, radar systems, the Littoral combat ships, simulation and training services; and unmanned systems and technologies. It also supports government customers in cybersecurity, and delivers communications, and command and control capabilities through mission solutions for defense applications. The Space segment provides satellites, strategic and defensive missile systems, and space transportation systems, and classified systems and services in support of national security systems. This segment also offers network-enabled situational awareness services and integrates space and ground-based systems to help its customers gather, analyze, and securely distribute critical intelligence data.

9.4 Teledyne Technologies Incorporated

1049 Camino Dos Rios

Thousand Oaks, CA 91360

United States

9.4.1 Company Basic Information

TABLE 51 TELEDYNE TECHNOLOGIES INCORPORATED BASIC INFORMATION AND ITS COMPETITORS

#	Item	Description
1	Company Name	Teledyne Technologies Incorporated
2	Website	www.teledyne.com
3	Primary Business	Scientific & Technical Instruments
4	Business Segments	Instrumentation, Digital Imaging, Aerospace and Defense Electronics, Engineered Systems
5	Key Executives	Al Pichelli - President and Chief Executive Officer
6	Key Strategies	Business Development
7	Competitors	Kongsberg Maritime, Oceaneering International, Inc., Lockheed Martin Corporation, Saab Group, Blueye, Aquabotix Technology Corporation, The Boeing Company, Fugro N.V., Subsea 7 S.A.

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

9.4.2 Teledyne Technologies Incorporated Product Category, Application, and Specification

9.4.2.1 Gavia Autonomous Underwater Vehicle (AUV)

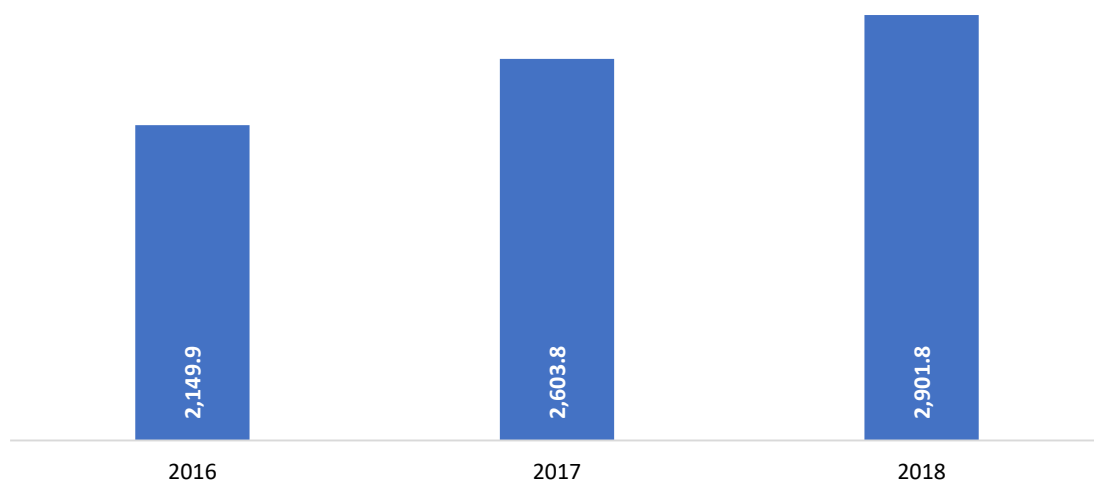
The Gavia is an Autonomous Underwater Vehicle that is made up of “plug-and-play” AUV modules which can be brought together and configured in the field. When assembled with a set of survey-grade sonar modules, a Gavia AUV becomes a self-contained survey solution with a low logistics footprint that is carrying out with a wide range of missions for commercial, defence, and scientific applications. Gavia modules are available in 500 or 1000m versions and are fully interchangeable on any Gavia AUV in the field.

9.4.2.2 SeaBotix Remotely Operated Vehicles (ROVs)

Remotely Operated Vehicles (ROVs) are powerful, compact, open-frame observation and inspection class vehicles with extensive built-in flexibility. SeaBotix ROVs have the advantage of being small and lightweight, yet powerful enough to navigate strong currents, carrying a multitude of sensors and payloads. This makes them ideally suited for tasks such as ship hull inspection, dam and tunnel inspection, and search and recovery operations.

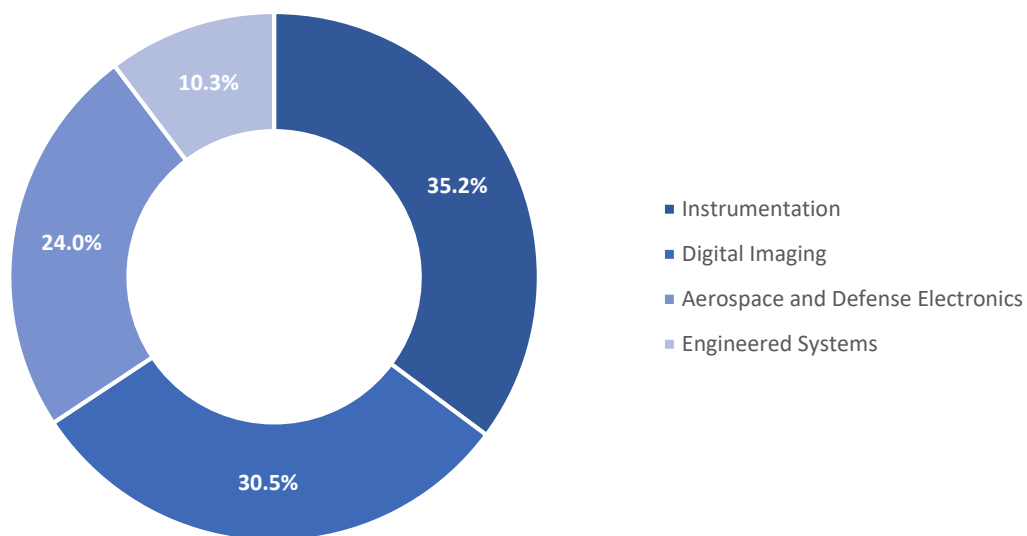
9.4.3 Teledyne Technologies Incorporated Financial Performance (2016-2018)

FIGURE 32 TELEDYNE TECHNOLOGIES INCORPORATED REVENUE, 2016-2018, (\$MILLION)



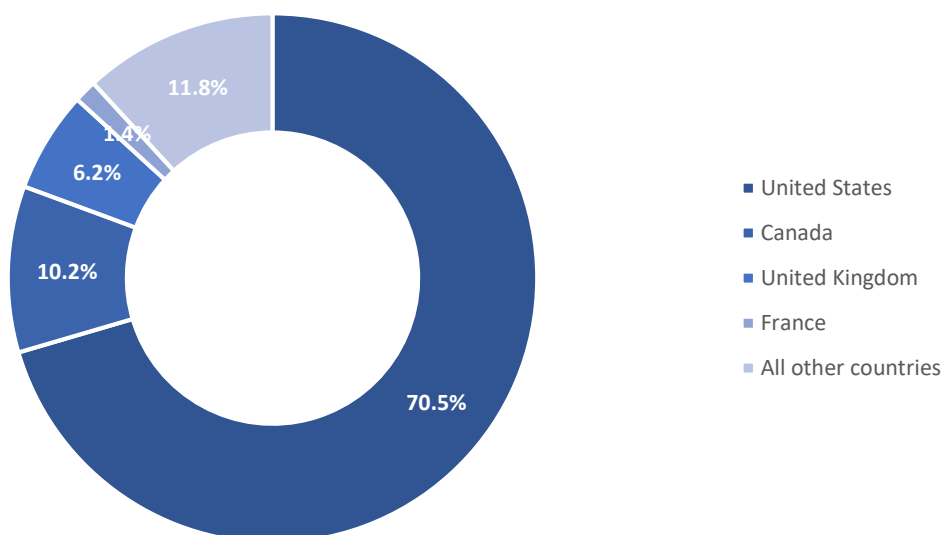
Source: Secondary Sources, Company Publications, Primary Interviews, and ERC Analysis

FIGURE 33 TELEDYNE TECHNOLOGIES INCORPORATED REVENUE, BY BUSINESS SEGEMENT, 2018 (%)



Source: Secondary Sources, Company Publications, Primary Interviews, and ERC Analysis

FIGURE 34 TELEDYNE TECHNOLOGIES INCORPORATED REVENUE, BY REGION, 2018 (%)



Source: Secondary Sources, Company Publications, Primary Interviews, and ERC Analysis

9.4.4 Teledyne Technologies Incorporated Recent Development

TABLE 52 RECENT DEVELOPMENTS

Date	Type	Description
August 21, 2019	Business Development	Teledyne Technologies Incorporated announced that its subsidiary, Teledyne Instruments, Inc., have received contract by the U.S. Naval Undersea Warfare Center to supply autonomous underwater vehicles (AUVs) and related monitoring and communications acoustic systems.

Source: Company publications & ERC Analysis

9.4.5 Company Overview

Teledyne Technologies Incorporated was founded in 1960 and is headquartered in Thousand Oaks, California. Teledyne Technologies Incorporated provides instrumentation, digital imaging, aerospace and defense electronics, and engineered systems in the United States, the United Kingdom, Denmark, Canada, France, and the Netherlands. The company's Instrumentation segment offers monitoring and control instruments for marine, environmental, industrial, and other applications, as well as electronic test and measurement equipment, and power and communications connectivity devices for distributed instrumentation systems and sensor networks deployed in mission critical and harsh environments. Its Digital Imaging segment provides image sensors and digital cameras for use in industrial, scientific, and medical applications, and hardware and software for image processing and automatic data collection in industrial and medical applications, as well as manufacturing services for micro electro-mechanical systems. This segment also offers light detection and ranging systems, focal plane arrays, sensors, and subsystems, and geospatial software products. The company's Aerospace and Defense Electronics segment provides electronic components and subsystems, as well as communications products, such as defense electronics, environment interconnects, data acquisition and communications equipment for aircraft, components and subsystems for wireless and satellite communications, and general aviation batteries. Its Engineered Systems segment offers systems engineering and integration, technology development, and manufacturing solutions for defense, space, environmental, and energy applications, and designs and manufactures electrochemical energy systems and small gas turbine engines. The company markets and sells its products and services through sales forces, third-party distributors, and commissioned sales representatives.

9.5 Saab AB

Olof Palmes gata 17, 5 tr

Stockholm, 111 22

Sweden

9.5.1 Company Basic Information

TABLE 53 SAAB AB BASIC INFORMATION AND ITS COMPETITORS

#	Item	Description
1	Company Name	Saab AB
2	Website	www.saabgroup.com
3	Primary Business	Radar and sensor technology
4	Business Segments	Aeronautics, Dynamics, Surveillance, Support and Services, Industrial Products and Services, Kockums, Corporate
5	Key Executives	Mr. Håkan Buskhe - CEO, President & Director
6	Key Strategies	Business Development
7	Competitors	Kongsberg Maritime, Oceaneering International, Inc., Lockheed Martin Corporation, Teledyne Technologies, Inc., Blueye, Aquabotix Technology Corporation, The Boeing Company, Fugro N.V., Subsea 7 S.A.

Source: Secondary Research, Company Publications, Primary Interviews, and ERC Analysis

9.5.2 Saab AB Product Category, Application, and Specification

9.5.2.1 Falcon

The Falcon is an underwater electric robotic system equipped with Saab Seaeye's advanced iCON intelligent control system. The Falcon provides exceptional vehicle control and diagnostic data as well as the ability to customize the pilot display and enable features such as station keeping.

9.5.2.2 Tiger

The Tiger is a very stable platform, with excellent maneuverability and performance in strong currents. Its open frame construction and generous payload offer the possibility of adding a wide range of tools and sensors as well as interchangeable tool skids. The Tiger is available as a free-swimming vehicle down to depths of 450m.

9.5.2.3 Lynx

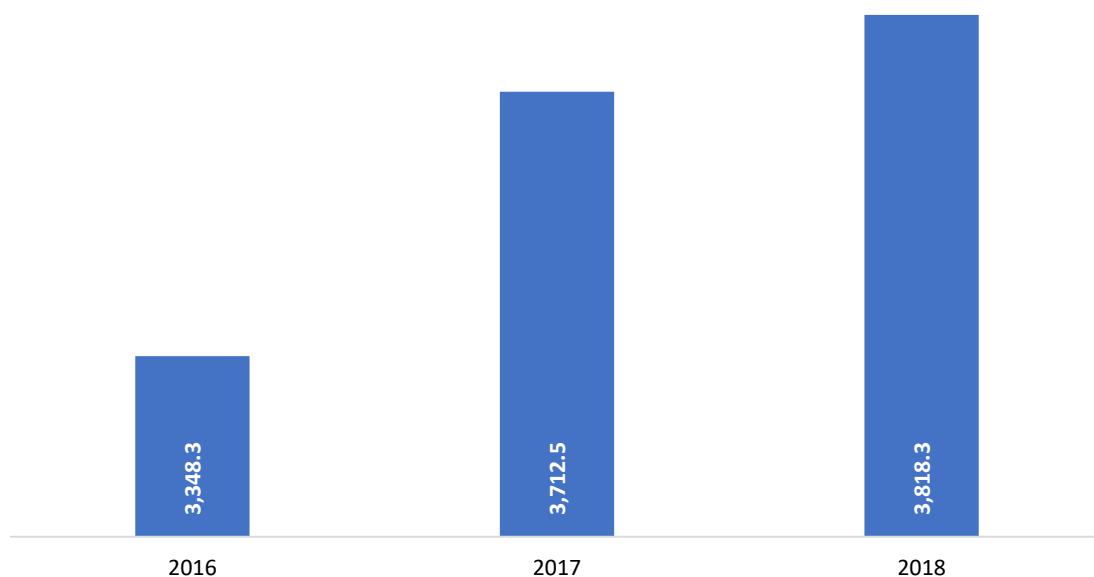
The Lynx is a leading observation and inspection vehicle used by the oil and gas industry especially for missions in water depths to 1,500m. The Lynx is fitted with two vertical and four horizontal thrusters making it a very stable platform with exceptional maneuverability.

9.5.2.4 Others

The others Saab's Underwater Vehicles includes Cougar-XT Compact, SeaOwl-XTi, Cougar-XT, Cougar-XTi, Panther-XT, Panther-XT Plus, Leopard, Jaguar, Sabertooth Single Hull, and Sabertooth Double Hull.

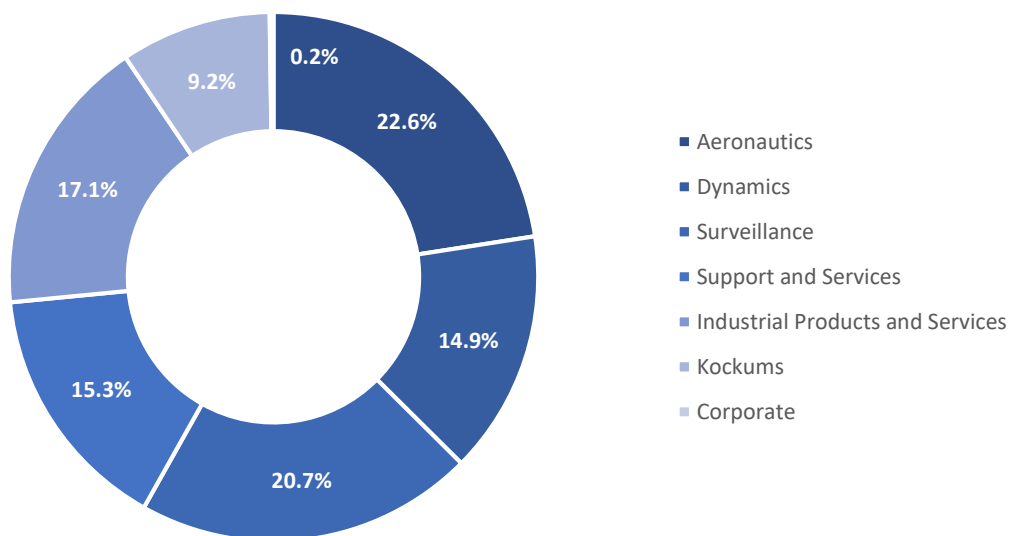
9.5.3 Saab AB Financial Performance (2016-2018)

FIGURE 35 SAAB AB REVENUE, 2016-2018, (\$MILLION)



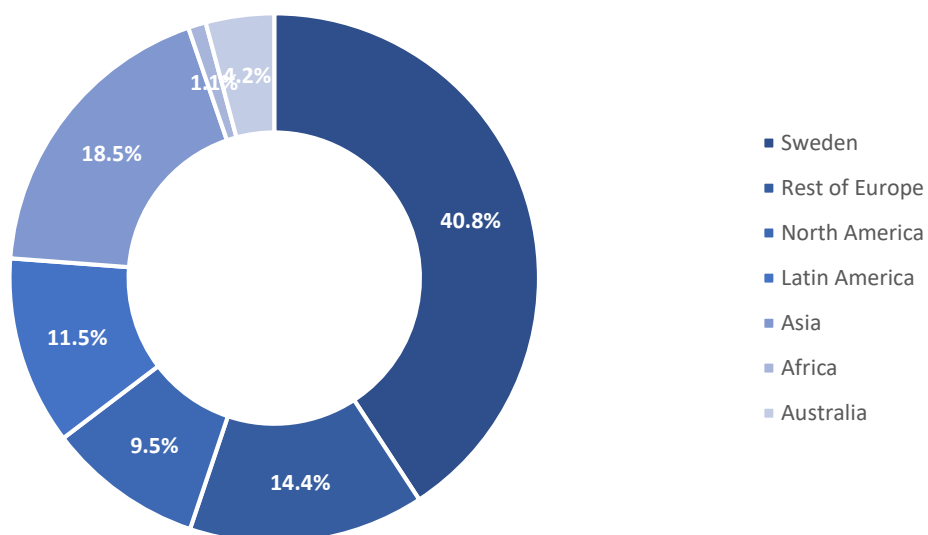
Source: Secondary Sources, Company Publications, Primary Interviews, and ERC Analysis

FIGURE 36 SAAB AB REVENUE, BY BUSINESS SEGEMENT, 2018 (%)



Source: Secondary Sources, Company Publications, Primary Interviews, and ERC Analysis

FIGURE 37 SAAB AB REVENUE, BY REGION, 2018 (%)



Source: Secondary Sources, Company Publications, Primary Interviews, and ERC Analysis

9.5.4 Saab AB Recent Development

TABLE 54 RECENT DEVELOPMENTS

Date	Type	Description
May 14, 2019	Business Development	Saab have received order in collaboration with the UK-based company QinetiQ. The order comprises of Saab's AUV62-AT autonomous underwater vehicle system. This system is a cost-efficient training solution allowing ASW units to train in a wide range of scenarios.

Source: Company publications & ERC Analysis

9.5.5 Company Overview

Saab AB was founded in 1937 and is headquartered in Stockholm, Sweden. Saab AB provides products, services, and solutions for the military, defense, and civil security markets worldwide. The company develops military and civil aviation technology, as well as conducts studies on manned and unmanned aircraft systems. The company also provides ground combat weapons, missile systems, torpedoes, unmanned underwater vehicles, training and simulation systems, and signature management systems for armed forces, and niche products for the civil and defense market, such as underwater vehicles for the offshore industry. In addition, the company offers solutions for safety and security, surveillance and decision support, and threat detection and protection, which include airborne, ground-based and naval radar, electronic warfare, and combat systems, as well as C4I solutions. Further, it provides product integration, support, technical maintenance, and logistics services, as well as products, solutions, and services for military and civil missions. Additionally, the company offers technology for traffic management and public safety, and supports industry, defense, and public sectors with services comprising cyber security, digitalization, and R&D. It also provides design, construction, and in-service support services for naval systems, such as surface combatants and submarines to the Swedish armed forces and other customers, and air-independent Stirling systems, mine countermeasure systems, and submarine rescue systems.

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